

What Are Empirical Monetary Policy Shocks? Estimating the Term Structure of Policy News*

Jonathan J. Adams[†] Philip Barrett[‡]
Federal Reserve Bank of Kansas City International Monetary Fund

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Abstract

Empirical monetary policy shocks (EMPS) mix information about both current and future policy. Policy news shocks at different horizons have different macroeconomic effects, so quantifying this mix is essential to use EMPS to evaluate theory. To disentangle these shocks, we develop an IV method to estimate the *term structure of monetary policy news*, which captures how an EMPS affects policy residuals at each future horizon. Applying our method to popular monetary policy shocks, we learn that they do not represent textbook surprises. Instead, they mix information about policy at many horizons, and this mix varies depending on how the EMPS is identified. We can use the estimated term structures to construct synthetic shocks with arbitrary term structures and assess their macroeconomic effects. Synthetic surprise interest rate hikes are contractionary with little effect on prices, while long-term forward guidance is deflationary.

JEL-Codes: E32, E43, E52

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[†]email: adamsjonathanj@gmail.com, website: jonathanjadams.com

[‡]email: pbarrett@imf.org, website: sites.google.com/view/philip-barrett

1 Introduction

How do central bank decisions affect the economy? Empirical answers to this question require well-identified monetary policy shocks. In recent years, clean identification using high-frequency data (Kuttner, 2001; Gürkaynak et al., 2005) or narrative methods (Romer and Romer, 2004) have yielded an array of high quality empirical monetary policy shocks (EMPS).

But what are these EMPS? The premise of this paper is that these approaches correctly capture the nature of the shock – i.e. an exogenous perturbation to interest rate policy – but they may vary in their information about policy timing. That is, EMPS do not exactly represent the textbook monetary policy shock, which is typically implemented immediately and without warning (i.e. a monetary policy surprise). Instead, each EMPS embodies a different combination of both policy surprises *and* news about future policies (i.e. forward guidance shocks).¹

Taking this idea seriously poses a challenge when trying to confront theory with data. If models imply that shocks with news at different horizons should have different effects, but EMPS are some combination of shocks at different horizons, then how should we interpret estimated responses to EMPS? Does the response to a given EMPS tell us something about the macroeconomic effects of policy? Or just how news and surprise are combined in that particular empirical shock? Without a way to put some discipline on how a given EMPS combines shocks at different horizons, it is impossible to say.

Our first contribution is to resolve these questions by developing a method to estimate the *term structure of monetary policy news*. This term structure decomposes an EMPS, revealing how it depends on policy surprises and news shocks for every future horizon. Our method to estimate the term structure has several stages. In the end, the estimator has a single closed-form expression, but it is helpful to describe it in distinct steps. First, we use plausibly exogenous macroeconomic shocks as instrumental variables in order to

¹This conundrum is well-known. Their creators often emphasize that EMPS are not pure monetary surprises; for example Gertler and Karadi (2015) describe their shock as “a linear combination of exogenous shocks to the current and expected future path of future rates.” Swanson (2021) describes the challenge thus: “identifying the effects of forward guidance and [large scale asset purchases] is difficult, because many of the FOMC’s announcements provide information about both types of policies simultaneously”.

identify the monetary policy rule, following insights from Barnichon and Mesters (2020).²³ Next, the monetary policy residual is calculated from the estimated rule, and then stripped of correlation with lagged residuals to find the monetary policy innovations. Finally, the innovations are regressed on lags of the EMPS to identify the term structure. It is possible to implement these steps as a single estimator with a simple closed-form expression; we prove that it consistently estimates the term structure.

Our second contribution is to estimate the term structure of sixteen well-known EMPS. We find that no EMPS is exactly a textbook monetary policy surprise. Even those with the shortest term structures include some information several months ahead. For example, narrative-based EMPS such as that of Aruoba and Drechsel (2024) come closest to a policy surprise, yet they still contain significant information at horizons up to six months. In contrast, EMPS intending to capture forward guidance do so effectively; they have nearly zero news about very short-run policy. But there remains heterogeneity depending on the identification strategy: the Swanson (2024) forward guidance shock is concentrated on policy six months to a year in the future; news in the Jarociński (2024) forward guidance shock is more evenly spread across two to 24 months; and the Bu et al. (2021) shock is almost entirely about horizons beyond one year. But if no empirical shock corresponds exactly to the objects in theoretical models, how can we compare the empirical and theoretical effects of monetary policy shocks? Our third contribution provides an answer.

Third, we develop a method to construct *synthetic monetary policy shocks* with any desired term structure from a linear combination of extant EMPS. This permits the construction of synthetic shocks that closely approximates a true policy surprise, news about a particular horizon, or any other pattern of forward guidance. We apply this method by estimating the effects of a synthetic surprise constructed from a linear combination of five recent EMPS and find that, as is usually expected, an interest rate surprise raises rates and causes a contraction in real activity. However prices have near zero response. In contrast, longer-run forward guidance shocks feature a small output contraction but a relatively large deflationary effect. We thus conclude that the observed deflationary effects of many of the component EMPS are mainly due to the contributions of long-run forward guidance. De-

²³This is a crucial step if there is a non-trivial news component to monetary policy. Many studies use lagged macroeconomic aggregates as instrumental variables to estimate the policy rule, following Clarida et al. (2000). These are valid instruments if monetary policy residuals are unanticipated, but may not be valid instruments if the policy residual is not a true surprise. The central insight of Barnichon and Mesters (2020) is that externally-identified exogenous macroeconomic shocks can be used to identify structural equations when the shocks are orthogonal to the relevant residual. Carvalho and Nechio (2014) argue that instruments may not be needed at all, and OLS estimates are reasonably accurate; as a robustness check, we use OLS to estimate the policy rule as well. While OLS may be preferable to IV using traditional lagged macro variables (Carvalho et al., 2021), we find IV using structural shocks to be more robust.

³Other examples applying this idea include: Caldara and Kamps (2017) use this approach to estimate a fiscal rule; Holden (2024) uses oil shocks to instrument for inflation in a policy rule for the real rate.

composing EMPS into their term structures in this way reveals otherwise hidden features of the underlying mechanisms.

Our construction of synthetic MPS has connections to the Lucas critique-robust policy counterfactual method developed by McKay and Wolf (2023). McKay and Wolf use a linear combination of EMPS to construct counterfactual IRFs to non-monetary shocks. In contrast, we use a linear combination of EMPS to construct a monetary shock with a synthetic term structure of policy news. And the connection goes further, because our method is valuable for implementing a prescribed counterfactual policy in practice. Fundamentally, the McKay-Wolf method solves for how observed EMPS should contemporaneously respond to macroeconomic shocks to maximize some welfare criterion. But how would a central bank actually implement such a response? The bank has to be able to create EMPS at will, but this is not feasible without knowing what the EMPS are. The bank needs to know: How do the EMPS depend on interest rate actions today versus in the future? Our method answers this question.⁴

Aside from the central contributions of our paper, we make two additional technical advances. First, we demonstrate the benefits of contemporaneous IV estimation for monetary policy rules, which has been shown previously for fiscal policy rules (Caldara and Kamps, 2017). Our policy rule coefficients are surprisingly robust across specifications, and roughly match standard theoretical values (e.g. the inflation coefficient is $\phi_\pi \approx 1.5$). OLS is known to have only a small bias for estimating these rules (Carvalho et al., 2021), but is much more sensitive to the regression specification, compared to the IV approach. Second, while deriving a penalized version of our estimator for finding smooth term structures, we utilize the Barnichon and Brownlees (2019) “smooth local projections”. The estimator is originally written non-linearly. Instead, we show how to rewrite the smooth local projection as a special case of ridge regression.

We contribute to the literature working to separately estimate the effects of forward guidance (news) versus policy surprises. Gürkaynak et al. (2005) decomposes high-frequency MPS into a target factor that moves the current rate, and a path factor that only moves expected future rates. Other papers such as Altavilla et al. (2019), Swanson (2021), and Jarociński (2024) decompose high-frequency shocks into additional factors, which have different macroeconomic effects. We show in Section 4 that the shocks resulting from these decompositions are characterized by different news term structures. Campbell et al. (2012) estimate a simple Taylor rule, and use forecasts to decompose the residual into components revealed when the rate is set versus in prior quarters. Hansen and McMahon (2016) use textual analysis to identify components of Fed announcements corresponding to current

⁴This challenge and resolution also apply to the real-time optimal policy estimates derived by Barnichon and Mesters (2023).

policy, views about the economy, and forward guidance. Many further papers apply these types of strategies to other settings.⁵

The remainder of the paper is organized as follows. Section 2 contains a motivating example to demonstrate why knowing the term structure of an EMPS is necessary to draw conclusions. Already-motivated readers can skip to Section 3, which describes our method in detail. In Section 4 we apply it to estimate the term structures for many EMPS. Section 5 describes and applies the process for constructing synthetic MPS. We present robustness checks in Section 6. Section 7 concludes.

2 A Motivating Example

Our motivation is most clearly demonstrated with a concrete example. In this section, we show that for almost all models there is some term structure which can rationalize any given EMPS. Consequently, without some empirical discipline on the term structure of an EMPS, we cannot use them to evaluate theory.

The textbook New Keynesian model is given by

$$\begin{aligned} \text{New Keynesian Phillips curve:} \quad & \pi_t = \beta \mathbb{E}[\pi_{t+1}] + \kappa y_t \\ \text{Euler equation:} \quad & i_t = \mathbb{E}_t[\gamma(y_{t+1} - y_t) + \pi_{t+1}] \\ \text{Taylor rule:} \quad & i_t = \phi_y y_t + \phi_\pi \pi_t + \nu_t \end{aligned}$$

where π_t is inflation, y_t is the output gap, and i_t is the nominal interest rate. ν_t is exogenous and white noise. However, we introduce news to this model, similar to how McKay and Wolf (2023) do so: ν_t is partially anticipated, given by

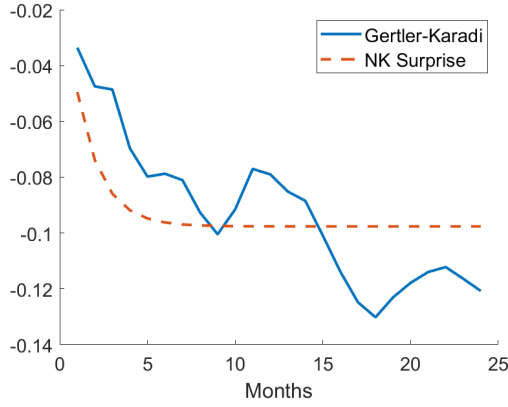
$$\nu_t = \nu_{0,t} + \nu_{1,t-1} + \nu_{2,t-2} + \dots$$

where the component $\nu_{h,t-h}$ is learned at time $t - h$. The $\nu_{h,t}$ components are i.i.d. over time and independent of one another. $\nu_{h,t}$ represents a news shock at time t about monetary policy h periods into the future.

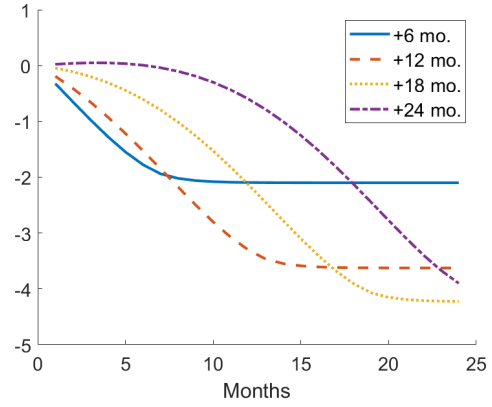
Figure 1a compares the price level IRFs from the New Keynesian model to that of a well-known high-frequency EMPS, that of Gertler and Karadi (2015). The shock causes a gradual deflation over 18 months. In contrast, the standard New Keynesian monetary policy surprise $\nu_{0,t}$ (dashed red line) causes an immediate deflation, then prices rapidly stabilize.

But a surprise is not the only kind of monetary policy shock. A news shock $\nu_{h,t}$ has

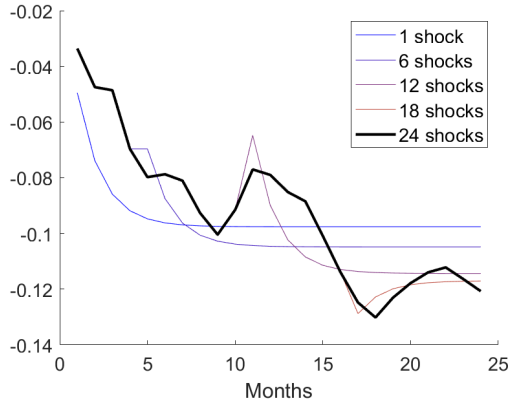
⁵Fawley and Neely (2014) document the history of the identification of monetary policy shocks.



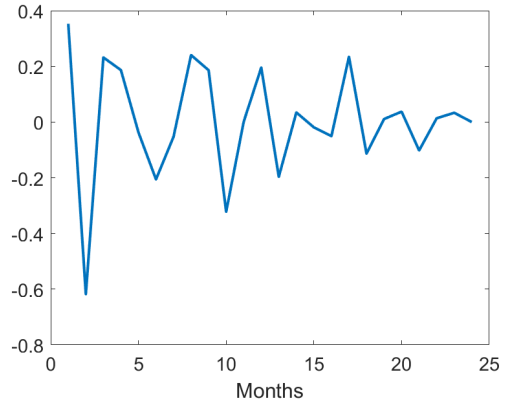
(a) EMPS vs. NK Surprise Shock



(b) NK News Shocks



(c) MPS Approximations



(d) Rationalizing Term Structure

Figure 1: CPI Responses to Monetary Shocks

The EMPS IRF is directly from Gertler and Karadi (2015). IRFs to surprise and news shocks are calculated from a standard calibration (Galí, 2008) of the textbook New Keynesian model. The MPS Approximation IRFs use the first n news shocks to find the linear combination that most closely matches (in terms of least squares) the EMPS IRF. The Rationalizing Term Structure plots the weights on the news shocks that exactly recover the Gertler-Karadi IRF.

a different effect on prices for every horizon h : an anticipated future tightening causes a smooth deflation. Panel 1b demonstrates this by plotting the deflationary effects of news at several semi-annual horizons. Each looks different from a surprise shock, and different from one another. Indeed, they are linearly independent.

The IRF of the Gertler-Karadi shock is perfectly consistent with the New Keynesian model for *some* term structure. In other words, there is some linear combination of surprise and news that exactly replicates the empirical IRF. Panel 1c demonstrates, by approximating the Gertler-Karadi IRF as linear combinations of the first n news horizons. As n increases, the IRF is approximated more accurately. When 24 shocks are used, the Gertler-Karadi IRF is reproduced perfectly. Panel 1d plots the weights on each news shock in the perfect approximation: this linear combination generates an MPS that would exactly rationalize the Gertler-Karadi IRF in the textbook New Keynesian model.⁶ Moreover, if appropriately rescaled, this is the term structure of monetary policy news, which we define formally in the next section.⁷

The lessons from this example are not limited to the Gertler-Karadi EMPS or the basic New Keynesian model. If the term structure of an EMPS is a free variable, we could have argued that any other model (with linearly independent news shock IRFs) was consistent with this EMPS for some term structure. Similarly, we could have found a term structure to rationalize any other EMPS as consistent with the New Keynesian model. The essential point is that without some discipline on the term structure of monetary policy news, *anything goes*.

3 Methodology

This section describes the methodology used to estimate the term structure of monetary policy news. We outline the monetary policy framework, the estimation strategy, and the theoretical properties of the estimator.

3.1 Monetary Policy Framework

We model monetary policy as being determined by a Taylor-type rule:

$$y_t = x_t\phi + r_t \tag{1}$$

⁶Of course, one could use responses of additional variables to try to distinguish between models. But with enough news horizons, the problem returns. One can fully explain the H -period IRF for N variables with monetary shocks up to NH periods ahead.

⁷It is not smooth of course, as all news shocks have smooth IRFs in the NK model, so jagged linear weights are required to recover the jagged Gertler-Karadi IRF. This also prompts the question: How close can a smooth term structure come to matching the empirical IRF? Is there a tradeoff between smoothness of the term structure and matching the IRF? We return to these questions in Section 3.

where y_t is the policy instrument (typically a short-term rate), r_t is the exogenous monetary policy residual, x_t is a row vector of n_x endogenous inputs to the policy rule, and ϕ is a vector of coefficients. Equation (1) represents a linear approximation of the possibly nonlinear true function mapping the state of the economy to policy decisions.⁸

We allow the residual r_t to be autocorrelated. Its data-generating process is

$$r_t = \sum_{\ell=1}^L \rho_{r,\ell} r_{t-\ell} + \nu_t \quad (2)$$

An autocorrelated residual is one way to model interest rate smoothing. Alternatively, one might forego the use of equation (2) by including sufficient lags of interest rates and observables in the x_t vector. But splitting them out maintains the familiarity and parsimony of a Taylor-type rule in equation (1), while allowing for autocorrelated residuals. Because there are multiple observationally equivalent ways to model equations (1) and (2), we focus on the monetary policy innovation ν_t instead of a particular residual r_t .

The monetary policy innovation ν_t is white noise, but not necessarily unforecastable. We write the residual ν_t as a sum of news shocks at H_ν horizons:

$$\nu_t = \nu_{0,t} + \nu_{1,t-1} + \nu_{2,t-2} + \dots + \nu_{H_\nu,t-H_\nu} \quad (3)$$

$\nu_{0,t}$ represents the surprise at time t , while $\nu_{h,t-h}$ represents the news component known at time $t - h$. This captures the idea that there may be information today about how policymakers intend to depart from their usual behavior in future. The news shocks are independent Gaussians, distributed $\nu_{h,t} \sim N(0, \sigma_h^2)$.⁹ Note that the assumption that the $\nu_{h,t-h}$ are independent structural shocks does not rule out correlated information at different horizons. For example, when the Fed signals tighter policy at one horizon, it will typically also signal tighter policy at other horizons. In our framework, such a tendency is captured by the autocorrelation structure in equation (2).

We model an EMPS as containing some information about news shocks at multiple horizons. There may be many types of EMPS, indexed by $j \in \mathcal{J}$. Each EMPS w_t^j contains information about potentially many future residuals, as well as Gaussian error ξ_t :

$$w_t^j = \sum_{h=0}^{H_w} \beta_h^j \nu_{h,t} + \xi_t^j \quad (4)$$

⁸So long as deviations are not too large, time-varying or state-dependent policies rules are allowed because, to first-order, the relevant state or time variables can appear in x_t .

⁹We assume Gaussianity so that we can write linear projections as expectations. This assumption is not necessary for our results; without it, the regression implementation would be unchanged.

where ξ_t is orthogonal to the monetary policy innovation ν_{t+h} for all h . ξ_t could be measurement error, but it could also represent other factors captured in the EMPS which do not affect the policy residual, such as a central bank information effect. Equation (4) represents the data-generating process for an EMPS.

We define the *term structure of EMPS* j as the effect of the EMPS w_t^j on expectations of the monetary policy innovation ν_t over many horizons:

$$\gamma_h^j \equiv \frac{d\mathbb{E}[\nu_{t+h}|w_t^j]}{dw_t^j}$$

Given the linear DGP in equation (4), the term structure can also be written as a linear relationship between EMPS w_t^j and ν_t :

$$\nu_t = \sum_{h=0}^{H_w} \gamma_h^j w_{t-h}^j + u_t^j \quad (5)$$

where u_t^j is an orthogonal residual. The β_h^j coefficients from equation (4) and γ_h^j coefficients are related by

$$\gamma_h^j = \beta_h^j \frac{Var(\nu_{h,t})}{Var(w_t^j)} \quad (6)$$

Whether the user is ultimately interested in the γ_h^j or β_h^j representation, the two are fundamentally related by equation (6).¹⁰ However, the value of β_h^j will not be recoverable from γ_h^j without knowing $Var(\nu_{h,t})$ which is typically unobserved.

Equation (5) encodes the term structure, but cannot be directly estimated since the EMPS w_t^j are data, but the monetary policy innovations ν_t are not. The next section describes how to estimate the term structure using instrumental variables.

3.2 Estimation Strategy

Estimating the γ_h^j coefficients from equation (5) faces several challenges: ν_t is unobserved, it is unknown how it relates to the monetary policy residual r_t , and r_t itself is not orthogonal to the endogenous variables x_t .¹¹

To resolve these challenges, our estimation takes a 4-stage approach. An important assumption in our method is the availability of a battery of other well-identified non-monetary

¹⁰And just as EMPS measurement error was allowed in equation (4), it is allowed when estimating the γ_h^j term structure as well. This is because measurement error simply increases the common $Var(w_t^j)$ denominator in (6), and in our application we will only be concerned with the γ_h^j coefficients up to scale.

¹¹For example, this endogeneity is why the immediate response of the yield curve is not enough to infer the term structure of monetary policy news. In general, a shock at any single horizon can affect the entire yield curve, both at shorter and longer horizons: immediate policy shocks will affect future interest rates through internal propagation, while forward guidance shocks affect short-run interest rates due to anticipation effects.

macro shocks, z_t . We discuss these further in Section 4.1 but for now we take their existence as given. The steps are:

1. Instrument for the endogenous variables x_t in the policy rule, using exogenous macroeconomic shocks z_t that are orthogonal to both u_t and the monetary policy shocks w_t^j .
2. Use the instrumented variables to estimate the policy rule coefficients $\hat{\phi}$ from equation (1). This is standard 2SLS estimation.
3. Calculate the implied residuals $\hat{r}_t$ using the estimated policy coefficients $\hat{\phi}$:

$$\hat{r}_t = y_t - x_t \hat{\phi} \quad (7)$$

then whiten to find the estimated $\hat{\nu}_t$ innovations. In this step, we can project the residual $\hat{r}_t$ onto its own lags:

$$\hat{r}_t = \sum_{\ell=1}^L \hat{r}_{t-\ell} \varrho_\ell + \nu_t \quad (8)$$

4. Use the estimated $\hat{\nu}_t$ innovations to estimate the term structure γ_h^j of EMPS j from equation (5).

3.3 The Estimator

The 4-stage approach for estimating the γ_h^j coefficients is convenient because it is linear, and there is a closed form expression for the estimator. Proposition 1 gives the expression using the following notation. Let $\hat{\mathbf{r}}_{t-1} = (y_{t-1} - x_{t-1} \hat{\phi} \quad \dots \quad y_{t-L} - x_{t-L} \hat{\phi})$ denote the row vector of lagged estimated Taylor residuals. This allows us to write the whitening regression (8) as

$$\hat{r}_t = \hat{\mathbf{r}}_{t-1} \varrho + \nu_t \quad (9)$$

Similarly, we stack lags of EMPS in the vector $\mathbf{w}_t \equiv (w_t^j \quad w_{t-1}^j \quad \dots \quad w_{t-H_w}^j)$ which allows us to write the fourth regression (5) as

$$\hat{\nu}_t = \mathbf{w}_t \gamma + u_t \quad (10)$$

where we have suppressed the j superscript for readability.

To cleanly derive the estimator, we define X , Z , and $\mathbf{W}$ as matrices of the endogenous variables x_t , instruments z_t , and EMPS $\mathbf{w}_t$, respectively. Each row corresponds to a time t observation. y and u are vectors of policy observations and equation (5) residuals, respectively. The vector of estimated residuals is $\hat{R} = y - X \hat{\phi}$, and the matrix $\hat{\mathbf{R}}$ is constructed

by

$$\hat{\mathbf{R}} = \mathbf{Y} - \mathbf{X} \left(I_{n_x} \otimes \hat{\phi} \right)$$

where $\mathbf{y}_{t-1} = (y_{t-1} \dots y_{t-L})$ and $\mathbf{x}_{t-1} = (x_{t-1} \dots x_{t-L})$ are row vectors of lagged observables, while $\mathbf{Y}$ and $\mathbf{X}$ denote the corresponding matrices of observations. Define the orthogonalization matrices $M_{\hat{\mathbf{R}}} \equiv I - \hat{\mathbf{R}}(\hat{\mathbf{R}}'\hat{\mathbf{R}})^{-1}\hat{\mathbf{R}}'$ and $M_{IV} \equiv I - X(X'P_ZX)^{-1}X'P_Z$, where $P_Z \equiv Z(Z'Z)^{-1}Z'$ denotes the matrix projecting onto the instruments.

Proposition 1 *The 4-stage estimator $\hat{\gamma}$ is given by*

$$\hat{\gamma} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'M_{\hat{\mathbf{R}}}M_{IV}y$$

Proof: Appendix A

The $\hat{\gamma}$ coefficient vector can be estimated by four independent OLS regressions or in one step, following Proposition 1.

Why do we need to estimate the Taylor rule by IV, instead of simply estimating how the EMPS affect future policy rates? The challenge we face is that all monetary policy shocks – be they surprise rate hikes or forward guidance – affect future policy rates through the endogenous term $x_{t+h}\phi$ in equation (1). The IV steps are necessary to isolate how EMPS affect the policy residual alone, distinct from the endogenous propagation channels.¹² An ancillary benefit of our method is that we get clean estimates of the Taylor rule coefficients, the $\hat{\phi}$. These are given by:

$$\hat{\phi} = (X'P_ZX)^{-1}X'P_Zy \quad (11)$$

The intuition is that by using a battery of non-monetary shocks we can identify the systematic part of the policy rule by isolating variation in x_t independent of monetary policy.

What must be true for this estimation strategy to work? First, we need to correctly specify the policy rule. The rule represents the systematic, average way that policy decisions depend on the state of the economy, while the residual represents a transient deviation by monetary policymakers. Whether or not the rule is perfectly specified, our method needs to accurately isolate the innovations to these residuals. Thus, evaluating the variables we include in equation (1) and the lag length in equation (8) will constitute important robustness checks. Second, we require valid instruments to use in the IV step, which we discuss next. Our Monte Carlo simulation exercise, presented in detail in Appendix F, also touches on this issue, since it presents a specific case where the estimation strategy works even with weak instruments. The Monte Carlo simulation offers something of an end-run around many potential critiques of our approach. That this approach recovers the correct

¹²Appendix C further explores the quantitative importance of accounting for endogenous policy responses.

term structure of monetary shocks in the most common contemporary monetary model, is at least strongly suggestive that it should also work in practice.

We prove that if the macroeconomic shocks are valid instruments, then the 4-stage estimation approach is unbiased. We say that a time series of shocks z_t are valid instruments for estimating the policy rule (1) if they satisfy the usual properties: $\mathbb{E}[z_t r_t] = 0$ (exclusion restriction), $\mathbb{E}[z_t' x_t]$ has full column rank (relevance). We assume throughout that standard regularity conditions hold (e.g. finite conditional second moments) so that the law of large numbers applies. The key concern is whether the exclusion restriction applies: it's not enough that instruments are orthogonal to the w_t^j EMPS, they must be orthogonal to the full monetary policy residual r_t . This requires whichever shocks are used as instruments to have been carefully identified.

Proposition 2 *If z_t is a vector of valid instruments for estimating the policy rule (1), then the 4-stage estimator is consistent:*

$$plim(\hat{\gamma}) = \gamma$$

Proof: Appendix A

3.4 Generalization with Smoothing

The final stage of the 4-stage estimator is effectively a local projection (Jorda, 2005) because the EMPS in equation (5) are orthogonal. Local projections have many appealing properties, including that they are unbiased, which allowed us to prove that the entire 4-stage estimator is unbiased (Proposition 2). However, local projection estimates have large errors. Li et al. (2024) show that penalized local projections perform very well; allowing for a small amount of bias can substantially shrink the estimator variance. When considering the bias-variance trade-off, one's objective would have to place almost no weight on minimizing variance in order to prefer unpenalized local projections.

Therefore, we generalize our 4-stage estimator to allow for a penalty to reduce estimator variance. Specifically, in the 4th stage, we estimate a “smooth local projection” (Barnichon and Brownlees, 2019), which approximates an IRF with a set of smooth basis functions. Besides its popularity, this is an appealing method because it can be represented as a ridge regression. This means that we can write the generalized 4-stage estimator in closed form.

Appendix B describes how to estimate the canonical smooth local projections as a standard ridge regression. In this appendix, Proposition 8 defines the appropriate penalty matrix $\mathbf{P}_B$. The penalty parameter λ controls the degree of smoothing, and is selected by cross-validation. Proposition 3 gives the generalized *smoothed 4-stage estimator*. We call

it “generalized”, because it nests the original 4-stage estimator (Proposition 1) when the penalty is set to $\lambda = 0$.

Proposition 3 *The smoothed 4-stage estimator $\hat{\gamma}_\lambda$ for penalty parameter λ is given by*

$$(W'W + \lambda \mathbf{P}_B)^{-1} W' M_{\hat{\mathbf{R}}} M_{IV} y$$

Proof: Appendix B.2

We use the smoothed 4-stage estimator throughout the following empirical work.

4 Estimated Term Structures

In this section, we estimate the term structures of popular EMPS using our proposed methodology. We first describe the data used for the estimation, including the different EMPS series and the macroeconomic instruments. Then we present the estimation results, providing a summary statistic to represent the relative importance of news for each EMPS and discuss the implications of our findings. Finally, we report a summary of the findings of a much more extensive validation exercise using simulated data from a standard New Keynesian model, reported in detail in Appendix F.

4.1 Data

Our method requires two types of data: monetary policy shocks from the literature, and other macroeconomic instruments and series used to estimate the policy rule.

4.1.1 Monetary Policy Shock Data

We estimate the term structure of monetary policy news for a variety of well-known EMPS. They are summarized in Table 1.

Many shock series rely on intraday data for identification, constructing instruments based on high-frequency changes in asset prices around FOMC announcements as a measure of monetary policy surprises. A classic example, Gertler and Karadi (2015) use 3-month-ahead federal funds futures rates. This horizon covers multiple FOMC meetings, and is interpreted as capturing both current rate decisions and forward guidance. Bauer and Swanson (2023) refines standard high-frequency methods by including additional policy events (e.g. speeches and press conferences) to the usual FOMC announcements to add observations, while also orthogonalizing with respect to high-frequency data to ensure that the EMPS series is unforecastable. Swanson (2024) applies these refinements to the Swanson (2021) methodology, which uses multiple asset prices to construct three distinct EMPS

Shock Source	Method	Notes	Range
Gertler and Karadi (2015)	HFI	30 min. window around FOMC decisions	1990:M1-2007:M12
Jarociński and Karadi (2020)	HFI	Pure monetary shock purged of Fed information	1990:M1-2016:M12
Bundick and Smith (2020)	HFI	2 shocks to term structure uncertainty	1994:M2-2019:M06
Miranda-Agrippino and Ricco (2021)	HFI	Orthogonalized w.r.t. Greenbook forecasts	1991:M1-2009:M12
Bu et al. (2021)	HFI	Alternative without intraday data	1994:M2-2024:M12
Bauer and Swanson (2023)	HFI	Orthogonalized w.r.t. financial data	1988:M2-2023:M12
Swanson (2024)	HFI	Decomposed into 3 types of EMPS	1988:M2-2023:M12
Jarociński (2024)	HFI	Decomposed into 4 types of EMPS	1990:M2-2024:M09
Acosta et al. (2025)	HFI	Announcements, minutes, and press conferences	1995:M2 - 2024:M12
Aruoba and Drechsel (2024)	Narrative	Natural language processing of Fed docs	1982:M10-2008:M10

Table 1: Monetary Policy Shocks

(the “target rate”, “forward guidance” and “large-scale asset purchases” (LSAP)) that correspond roughly to effects at short, medium, and long-term yields.

One concern with high-frequency EMPS is that it includes a “Fed information effect” (Romer and Romer, 2000; Nakamura and Steinsson, 2018) where the central bank reveals private information about the state of the economy, which is independent of its policy residuals. We include two EMPS series that attempt to isolate the information effects from true policy shocks. Jarociński and Karadi (2020) measure high-frequency changes in interest rates and stock prices, and use sign-restrictions to isolate information from policy shocks, assuming that information moves rates and stock prices in the same direction, while policy has opposite effects.¹³ Miranda-Agrippino and Ricco (2021) identify a pure policy shock by orthogonalizing the EMPS with respect to internal Fed forecasts.

We also use a shock identified with narrative methods. The classic Romer and Romer (2004) shock identifies policy actions motivated by the Fed’s policy stance, rather than reactions to contemporaneous economic data, by orthogonalizing with respect to internal forecasts. In a modern refinement, Aruoba and Drechsel (2024) incorporate substantially more information, via natural language processing of internal Fed documents. Then they orthogonalize interest rate changes with respect to both forecasts and the text-based time series.

4.1.2 Data for Estimating the Monetary Policy Rule

In our baseline method, we specify the monetary policy rule (1) with the Effective Federal Funds rate as the policy variable, and with employment and PCE inflation on the right-hand side.

¹³When testing the Jarociński-Karadi shock, we use their sign-restricted series, as opposed to the “poor man’s” approximation.

Shock Source	Method	Notes	Range
<i>Fiscal Shocks</i>			
Romer and Romer (2016)	Narrative	Social Security expansions	1951:M1-1991:M12
Fieldhouse et al. (2018)	Narrative	Government housing purchases	1952:M11-2014:M12
Phillot (2025)	HFI	Futures yields around Treasury announcements	1998:M10-2020:M01
<i>Oil Shocks</i>			
Känzig (2021)	HFI	Oil supply news	1975:M1-2023:M6
Baumeister and Hamilton (2019)	SVAR	Oil supply, consumption/inventory demand	1975:M2-2024:M3
<i>Other Shocks</i>			
Kim et al. (2025)	External	ACI severe weather shocks	1964:M4-2019:M5
Adams and Barrett (2024)	SVAR	Shocks to inflation expectations	1979:M1-2024:M5

Table 2: Structural Shock Instruments

To address endogeneity concerns in estimating the Taylor rule, we employ instrumental variables (IVs) drawn from the literature. Over the last decade, the collection of well-identified macroeconomic shocks has expanded substantially. However, our options are limited because we require monthly series. Still, we were able to collect six monthly instruments that represent a variety of shocks. They are summarized in Table 2.

Our first group of instruments is related to government spending and debt. We utilize the narrative measure of transfer payment shocks constructed by Romer and Romer (2016). This measure uses historical accounts of Social Security benefits to identify changes in transfer payments that are not a systematic response to macroeconomic conditions. To capture government spending shocks, we use the Fieldhouse et al. (2018) narrative instrument constructed from significant regulatory events impacting federal housing agency mortgage holdings. This series captures the ex ante impact of policy changes on the capacity of agencies to purchase mortgages. It focuses on non-cyclically motivated policy interventions by the federal government, excluding changes resulting from the agencies' regular response to market developments. These non-cyclically motivated policy shifts provide a source of exogenous variation in credit supply within the mortgage market. Lastly, we use the US debt supply shocks identified by Phillot (2025) using high frequency asset price data around Treasury auction announcements.

Our next group of instruments captures exogenous variations in the oil market. First, we use oil supply news shocks identified through high-frequency changes in oil futures prices around OPEC production announcements (Känzig, 2021). Second, we employ structural oil shocks identified from a structural VAR by Baumeister and Hamilton (2019). This approach distinguishes contemporaneous shocks to oil supply and shocks to oil demand, and, unlike other methods, does not require that there is no short-run response of oil supply to the

price.

We take severe weather shocks from the Actuaries Climate Index, a meteorological time series for severe weather. We take this series as exogenous, and use as shocks the statistical innovations calculated by Kim et al. (2025).

Finally, we use the Adams and Barrett (2024) inflation expectation shocks. This series is derived from a structural VAR that identifies exogenous shocks to inflation forecasts. To do so, the approach identifies the dimension of the VAR statistical innovation that causes survey forecasts to deviate from the rational expectation. In models where belief distortions are exogenous and stochastic, this method identifies the exogenous shock. The instruments' coverage is not balanced, so we treat shocks as zero whenever data are unavailable.¹⁴

4.2 Estimation Results

In this section, we present the estimated term structures of each EMPS both numerically and graphically.

4.2.1 Estimated Taylor Rules

This section describes the first 2 stages of our 4-stage estimator: estimating the Taylor Rule. We find that the use of structural shocks as IVs produces remarkably robust estimates for the inflation coefficient, especially compared to OLS approaches. Our estimated values are largely consistent with typical calibrations in theoretical models, with an inflation coefficient of roughly 1.5 across multiple specifications.

The results of the Taylor rule estimation are shown in Table 3. In most cases we specify the FFR to depend on currently monthly inflation and real activity, as well as lags of the Taylor rule residual. In the baseline specification (first column), we use two variables in x_t : inflation, for which our preferred measure is the 12-month growth rate in the PCE index; and activity, for which we use detrended private non-farm employment. These correspond directly to the two legs of the Fed's mandate: employment and price stability. We estimate the term structure up to two years after the shock, so $H_w = 24$. When whitening the monetary policy residuals, we orthogonalize with respect to six lags, setting $L = 6$.¹⁵ Because instruments can have persistent effects, we include six lags of the IVs. Table 3 also includes many alternative measures of real activity. As expected, the estimated coefficient on real activity is affected by whether we use Christiano-Fitzgerald

¹⁴Our baseline approach estimates the Taylor rule using data from 1975:M1 - 2020:M2, excluding the ZLB period, so the zeros choice mainly affects the narrative government spending shocks in recent years. In Appendix D we show that dropping these shocks entirely does not substantially change our estimates.

¹⁵We vary these choices in the robustness checks found in Section 6.

filtered real GDP,¹⁶ GDP growth, an alternative detrending method, industrial production, or unemployment affects. However, the inflation coefficients are largely unchanged, and generally satisfy the Taylor principle, that $\phi_\pi > 1$. We also include some specifications where we introduce additional variables. We model explicit interest rate smoothing by including the lagged interest rate (instead of implicitly subsuming smoothing into the autocorrelation of the residual); this reduces the coefficient magnitudes. Including the excess bond premium has little effect. In contrast, the introduction of inflation expectations (as measured either by the Michigan Survey or the Cleveland Fed) substantially changes the size of the inflation coefficient. This is not surprising, as expectations in the data are highly correlated with current inflation, and if the Fed responds to both similarly, rules with different coefficients might be almost observationally equivalent.

We also consider alternative inflation measures. These results are reported in Table 4. All 12-month inflation measures tend to satisfy the Taylor principle. When we use 1-month measures, we find smaller coefficients. And using core PCE, which the Federal Reserve considers a better indicator than headline PCE of medium-term inflationary pressures, even the one-month measure conforms to the Taylor principle.

In Appendix D, we report several more variations. First, we allow for alternative lag lengths. Second, we drop various IVs from our estimation to ensure that no single category in Table 2 is driving our results. And third, our baseline Taylor rule is estimated using data beginning in January 1975 and omits the zero-lower-bound (ZLB) and Covid periods, but we consider alternative choices. Our Taylor estimates appear robust to all of these checks, except for the inclusion of the ZLB period, which is unsurprising since policy rates are pinned to zero during this period, and thus invariant to macroeconomic conditions.

These regular results from the structural IV estimation contrast sharply with OLS estimates. OLS estimates from the literature vary considerably, and our findings are no different. We ran several OLS specifications, and the coefficient estimates are highly sensitive to specification choice. As an example, we also report in Appendix D OLS results with small differences in the lag structure, and found estimates that are highly dissimilar from each other, let alone our IV results. In contrast, our IV method produces results which are stable across multiple specifications and consistent with theory.

4.2.2 Estimated Term Structures

Figure 2 plots the estimated term structure of monetary policy news for each EMPS, using the smoothed 4-stage estimator.¹⁷ The further a term structure curve deviates from zero,

¹⁶We use monthly GDP estimated from a Kalman Smoother which matches the quarterly NIPA data and monthly consumption series (Barrett, 2025).

¹⁷We set the smoothing parameter to $\lambda = 30$. This was chosen by initially minimizing the rolling out-of-sample errors for each EMPS separately. That is, for each of a large set of values of λ we estimate $\hat{\gamma}_\lambda$

	Baseline	CF-filtered GDP	HP-filtered GDP	IP	Unemployment	Add EBP	Add π_t^c , Clev.	Add π_t^c , Mich.	GDP growth
12-month PCE Inflation, demeaned	1.517*** (0.097)	1.536*** (0.106)	1.528*** (0.102)	1.510*** (0.108)	1.187*** (0.098)	1.511*** (0.094)	-0.249 (0.160)	2.294*** (0.259)	1.541*** (0.107)
Private nonfarm payroll employment, departures from quadratic log trend, demeaned	0.370*** (0.036)					0.377*** (0.035)	-0.033 (0.036)	0.408*** (0.036)	
GDP growth, annualized, demeaned									0.031 (0.024)
GDP, CF-low-pass		0.697*** (0.233)							
GDP, HP-filtered			0.273 (0.288)						
Unemployment rate					0.265*** (0.024)				
Excess bond premium, demeaned						0.452 (0.291)			
Cleveland Fed inflation expectations, 1 year, demeaned							2.180*** (0.121)		
Industrial Production, CF low-pass filter				0.074 (0.104)					
Michigan inflation expectations, 1 year, demeaned								-1.170*** (0.371)	
Residual autocorrelation	0.95	0.95	0.96	0.96	0.96	0.96	0.89	0.92	0.95
R^2	0.58	0.38	0.45	0.46	0.49	0.59	0.79	0.56	0.46
Observations	569	569	569	569	569	569	485	533	569

Table 3: Estimated Taylor Rule Parameters: Different real variables

Table reports the estimated Taylor rule parameters from the second stage of the four-stage method using instrumental variables. Residual autocorrelation is the first order autocorrelation of the monetary policy residual, r_t . R^2 is calculated as the fraction of variance in the policy rate explained by the contemporaneous systematic part of monetary policy, $x_t\phi$, and so $1 - R^2$ is that explained by the monetary policy residual, r_t . Standard errors are reported in parentheses.

	Baseline	Lagged interest rate	Core PCE	CPI	Inf. only	1m Inf.	1m, Inf. only	Core PCE, 1m	CPI, 1m
12-month PCE Inflation, demeaned	1.517*** (0.097)	0.774*** (0.087)			1.525*** (0.108)				
Federal Funds Rate, lagged one month		0.343*** (0.027)							
Private nonfarm payroll employment, departures from quadratic log trend, demeaned	0.370*** (0.036)	-0.115** (0.051)	0.423*** (0.028)	0.343*** (0.037)		0.344*** (0.047)	0.371*** (0.047)	0.340*** (0.047)	
12-month Core PCE Inflation, demeaned			2.031*** (0.101)						
12-month CPI inflation, demeaned				1.125*** (0.088)					
1-month PCE Inflation, annualized, demeaned						0.598*** (0.083)	0.627*** (0.083)		
1-month Core PCE Inflation, annualized, demeaned							1.511*** (0.150)		
1-month CPI inflation, annualized, demeaned								0.372*** (0.061)	
Residual autocorrelation	0.95	0.95	0.95	0.96	0.96	0.89	0.88	0.61	0.93
R^2	0.58	0.70	0.64	0.54	0.46	0.34	0.20	0.42	0.29
Observations	569	569	569	569	569	569	569	569	569

Table 4: Estimated Taylor Rule Parameters: Different inflation measures

Table reports the estimated Taylor rule parameters from the second stage of the four-stage method using instrumental variables. Residual autocorrelation is the first order autocorrelation of the monetary policy residual, r_t . R^2 is calculated as the fraction of variance in the policy rate explained by the contemporaneous systematic part of monetary policy, $x_t\phi$, and so $1 - R^2$ is that explained by the monetary policy residual, r_t . Standard errors are reported in parentheses.

the more information the EMPS has about monetary policy at that horizon. The figure immediately reveals heterogeneity across the shocks. Some have spikes at low horizons, others have most of their weight in the middle, and most – but not all – decay to zero at long horizons.

It is striking that all shocks have a non-trivial term structure at horizons longer than $h = 0$. In other words, even the best identified shocks typically include information about both monetary surprises and forward guidance. For shocks derived from high-frequency methods, this might arise either because Fed announcements genuinely do include correlated information, or because the high-frequency variables used to inform the magnitude of the shock are inherently forward-looking.

When interpreting the term structures in Figure 2 it is helpful to have a summary statistic which represents the relative importance of news for a given EMPS. To do so, we use the R_k^2 statistic, which captures how much of the information in an EMPS is due to news at horizon k .

Proposition 4 *The share R_k^2 of variation in monetary policy innovation ν_t that is explained by an EMPS at horizon k is*

$$R_k^2 \equiv \frac{\text{Var}(\mathbb{E}[\nu_t | w_{t-k}^j])}{\text{Var}(\mathbb{E}[\nu_t | \{w_{t-h}^j\}_{h=0}^{H_w}])} = \frac{(\gamma_k^j)^2}{\sum_{h=0}^{H_w} (\gamma_h^j)^2}$$

Proof: Appendix A

Table 5 reports several of these statistics for each monetary policy shock, again calculated using the smoothed 4-stage estimator as in Figure 2. The first column is $R_{0,1}^2$, which—since monetary policy announcements can happen on any day during a month—we interpret as the share of the EMPS that is due to an “immediate” change in the monetary policy innovation. Table 5 illustrates several of our key findings. The most obvious is that the shocks which explicitly aim to identify surprises have a large share of their term structure variation at short horizons. Specifically, the EMPS that are most driven by the immediate horizons are the Swanson (2024) and Jarociński (2024) FFR shocks (both of which are “target level” shocks, identified by methods which split separate shocks to the future path of policy), and the Aruoba and Drechsel (2024) narrative shock. It is intuitive that the FFR

repeatedly on a series of extending subsets of the data, each beginning at the (same) sample start date but incrementing the end month by one for each element of the series. For each data subset (the minimum subset length is 10 years) we compute the out-of-sample errors on equation (10) for the first 12 months after the end date. We then choose the value of λ which minimizes the average error across the extending windows. This approach is analogous to cross-validation, in that it minimizes the out-of-sample errors, but it preserves the time series structure of the data. But because this results in terms structures which depend differently on smoothing across the different EMPS. And so for comparability we set a common value for λ in our baseline results, which is close to the average of the EMPS-specific optimal values. We consider alternate values of λ in the robustness checks found in Section 6.

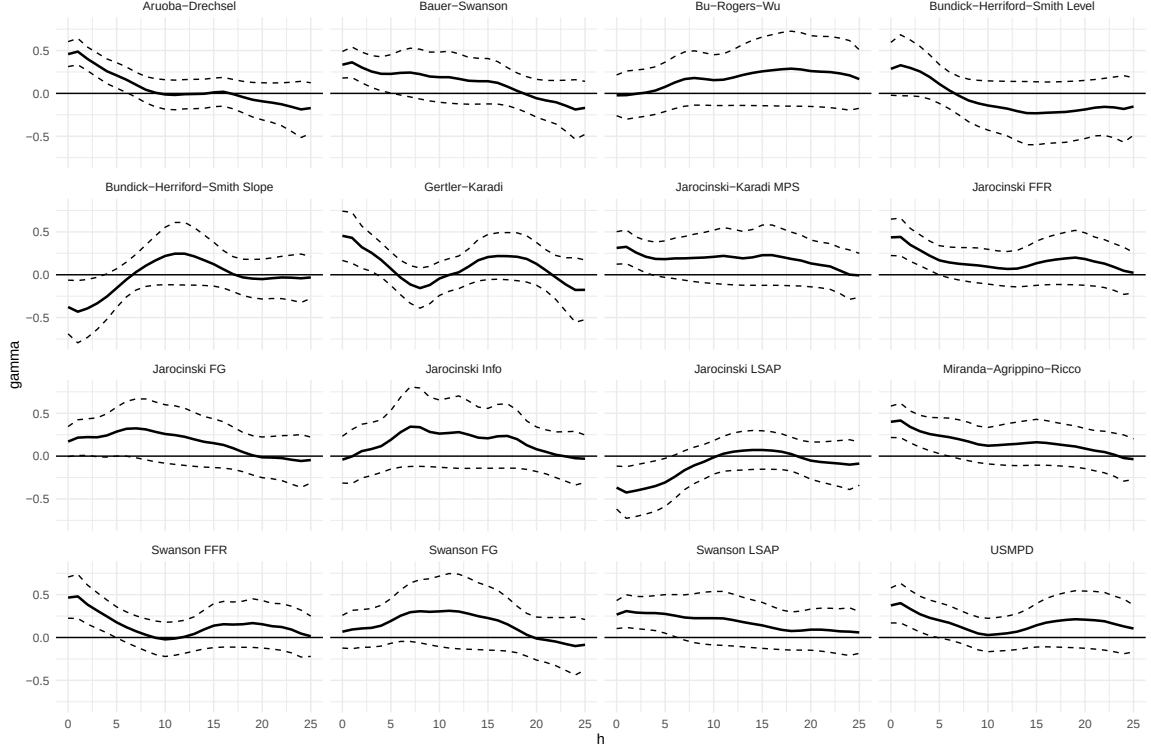


Figure 2: Estimated Term Structures

Figure shows the estimated γ_h^j coefficients, i.e. the impact of each identified monetary policy shock h periods later on the monetary policy innovation $\hat{\nu}_{t+h}$. The coefficients are normalized to have a unit squared sum. Term structures are smoothed using $\lambda = 30$. Dashed lines show bootstrapped 90%-10% confidence intervals.

shocks should appear here near the bottom of the table: these shocks are constructed by considering how many asset prices respond to Fed events, and isolating the only component that affects short term rates.¹⁸ However, no shock perfectly isolates true policy surprises. Even these well-constructed shocks which explicitly target interest rate surprises have much information at longer horizons. In some cases, this information dies out after around six months. But in other cases, the tail of the term structure is longer.

To summarize the forward guidance content of the EMPS, we calculate three components. The second column of Table 5 reports the sum of R_k^2 for $2 \leq k \leq 6$. This is *short-run news*, which is realized in the remaining half year after the immediate horizons. Column 3 reports *medium-run news*, which sums the R_k^2 statistic for next half year (months 7 – 12), and the final column reports *long-run news*, which occurs over the following year.

Reassuringly, two of the shocks that are the most driven by forward guidance are the

¹⁸Indeed, the estimated term structures assign even more short-term news to these FFR shocks than a naive regression on future policy rates. Appendix C explores how the term structures of policy news differ from the straightforward interest rate effects.

Shock	$R_{0:1}^2$	$R_{2:6}^2$	$R_{7:12}^2$	$R_{13:24}^2$
Swanson FG	0.004	0.113	0.575*	0.308***
Bu-Rogers-Wu	0.006	0.023	0.179*	0.792***
Jarocinski Info	0.016	0.121	0.553	0.310**
Jarocinski FG	0.058	0.323*	0.491*	0.128***
Swanson LSAP	0.180*	0.395**	0.298*	0.127**
Bundick-Herriford-Smith Level	0.224	0.230*	0.118	0.428**
Jarocinski-Karadi MPS	0.243	0.182*	0.233*	0.342**
Bauer-Swanson	0.274*	0.286*	0.247	0.192**
Bundick-Herriford-Smith Slope	0.308	0.319*	0.252	0.121**
Jarocinski LSAP	0.315	0.555**	0.039*	0.091**
USMPD	0.349*	0.284*	0.014	0.353**
Miranda-Agrippino-Ricco	0.395**	0.338**	0.106	0.161**
Gertler-Karadi	0.399	0.167*	0.095	0.339**
Jarocinski FFR	0.451**	0.256**	0.036	0.257**
Swanson FFR	0.489**	0.299**	0.010	0.202**
Aruoba-Drechsel	0.511***	0.379***	0.010	0.101*

Table 5: Decomposition of Term Structure by Horizon

Table reports the R_k^2 measures in Proposition 4, summed over monthly horizons denoted in subscripts. For example, $R_{2:6}^2$ is the total variation in the Taylor residual attributable to 2- to 6-month news in a given identified monetary policy shock. Stars denote whether 90% of the bootstrapped statistics exceed certain thresholds: * 0.025, ** 0.05, *** 0.10. Shocks are ordered by point estimates for $R_{0:1}^2$.

aptly-named Forward Guidance shocks estimated by Swanson and Jarociński. These shocks contain almost no information about policy surprises in the short-run. This of course does not have to be true; the shocks are constructed to have no information about policy surprises at very high frequencies, but could have a large $R_{0,1}^2$ statistic by containing extremely short-run news in the immediate or following month. But they do not. In Swanson’s case, the news is concentrated in the medium-run, while for Jarociński the news is more spread out over the two-year window. Joining these low-surprise examples is the Bu-Rogers-Wu shock, which is rare in that it is constructed to capture a high-frequency identification strategy without using proprietary high-frequency data. Their method produces a shock that overwhelmingly contains news about long-run policy; no other shock has more than 50% of its term structure appear more than a year in the future.

The R_k^2 statistics are also useful for clarifying uncertainty over information content. The estimated term structures reported in Figure 2 have in some cases large confidence intervals, computed via block bootstrap. But uncertainty over the point estimate of a single horizon in the term structure is not equivalent to uncertainty over whether the term structure is non-zero more broadly. To quantify uncertainty over the R_k^2 statistics (which are always positive), we calculate the 10th percentile of the bootstrapped statistics. Thus stars in Table 5 indicate whether 90% of the bootstrapped distribution exceed small, medium, or large thresholds. This quantification makes it clear that series such as Aruoba-Drechsel and the FFR shocks contain immediate surprises but also news about other horizons. And the forward guidance shocks are dominated by long-run news and nearly purged of short-run information.

4.3 Validation

To check our method, we also run a Monte Carlo exercise, testing our method on simulated data in small and large samples. Appendix F reports the results in detail, but the key findings are 1) that our method delivers unbiased estimates of both the Taylor rule coefficients and the term structure of EMPs in small samples, and 2) that the confidence intervals for the Taylor rule are accurate even with weak instruments for macroeconomic shocks. We also compare our results to applying OLS to the simulated data and show that the latter perform poorly in small samples. Although biases in Taylor rule estimation are economically small, confidence intervals are spuriously tight. The mapping from Taylor coefficients to the term structure is sensitive to this, leading to highly unreliable inference when using OLS estimates.

Our findings complement those of Carvalho et al. (2021), who find that the bias in OLS is small enough to be economically meaningless and, in small samples, preferable to traditionally-used GMM using lagged endogenous variables as instruments. One way to

interpret these results is to think of the different methods as picking different points on the trade-off between bias and maximizing informative variation. Lagged variable GMM following Clarida et al. (2000) reduces endogeneity bias (under certain assumptions) but throws away informative contemporaneous covariation of interest rates and macro variables. OLS exploits this variation but at the price of biased Taylor rule coefficients. Carvalho et al. (2021) show that this price is one typically worth paying in applied work. Our method gives the best of both worlds. It exploits contemporaneous variation in the endogenous variables, but by isolating only the variation due to non-monetary shocks it corrects for endogeneity bias. As a result, it delivers unbiased estimates of both the term structure and the Taylor rule coefficients, as well as accurate inference, even in small samples and, for the term structure at least, even with weak instruments.

5 Synthetic Monetary Policy Shocks

This section explains how to construct a synthetic monetary policy shock with a desired term structures, and then does so for several examples, including a synthetic surprise.

5.1 Method

The EMPS that we consider have a variety of news term structures. Calculating these term structures is innately useful, because it allows us to interpret the shocks in standard DSGE models. However, we can also use the results from multiple EMPS to construct *synthetic* shocks with a new term structure. This allows us to study the effects of MPS of particular interest that are not directly estimated in the data.

Let $\vec{\gamma}^j$ denote the vector of *normalized* term structure coefficients for EMPS j , estimated from Proposition 1, where the EMPS has been normalized so that $Var(w_t^j) = 1$.

Proposition 5 *For a linear combination of EMPS $w_t^c = \lambda_a w_t^a + \lambda_b w_t^b$, the resulting term structure of monetary policy news $\vec{\gamma}^c$ is proportional to the linear combination of term structures:*

$$\vec{\gamma}^c \propto \lambda_a \vec{\gamma}^a + \lambda_b \vec{\gamma}^b$$

Proof: Appendix A

Proposition 5 is useful because it allows us to construct a synthetic MPS with a desired term structure by finding the appropriate linear combination of existing EMPS. This is a valuable property because it allows us to study specific types of monetary policy shocks that are relevant to theoretical models but not directly estimated in the data. For example, one might be interested in studying the effects of a true monetary surprise, as in Figure 1a. But we learned in Section 4.2 that the EMPS all feature news at multiple horizons.

To estimate the effects of a surprise, we need to construct a synthetic MPS with a term structure $\bar{\gamma}^0 = \begin{pmatrix} 1 & 0 & 0 & \dots \end{pmatrix}'$. Or, if we wanted to study a pure 1-period-ahead news shock, we would construct a synthetic MPS with term structure $\bar{\gamma}^1 = \begin{pmatrix} 0 & 1 & 0 & \dots \end{pmatrix}'$. Indeed, the term structure of any h -period-ahead news shock is simply the corresponding basis vector. Proposition 6 states when this is feasible.

Proposition 6 *EMPS with normalized term structures in the set $\mathcal{J} = \{\bar{\gamma}^j\}$ can be used to construct any synthetic MPS s with term structure*

$$\bar{\gamma}^s \in \text{span}(\{\bar{\gamma}^j\}_{j \in \mathcal{J}})$$

This property follows directly from Proposition 5. An immediate corollary is:

Corollary 1 *If $\mathcal{J}$ contains $H_w + 1$ EMPS with linearly independent term structures, then a synthetic MPS can be constructed with any term structure of horizon length up to H_w .*

In practice, the number of linearly independent EMPS may be less than the IRF horizon $H_w + 1$. In this case, the span of the term structures is a lower-dimensional vector space. The synthetic MPS can be constructed with any term structure in that space. If the term structure of interest (e.g. $\bar{\gamma}^0$) is not in this space, it must be approximated. The following Proposition explains how to do so.

Proposition 7 *Let $\Gamma_{\mathcal{J}}$ denote the matrix of normalized term structures for the linearly independent set $\mathcal{J}$ of observed EMPS, and let $\bar{\gamma}^i$ denote the term structure of interest. The term structure of the synthetic MPS $\bar{\gamma}^s$ that is closest to $\bar{\gamma}^i$ (in the Euclidean norm) is given by*

$$\bar{\gamma}^s = \Gamma_{\mathcal{J}}(\Gamma'_{\mathcal{J}}\Gamma_{\mathcal{J}})^{-1}\Gamma'_{\mathcal{J}}\bar{\gamma}^i$$

Proof: Appendix A

In the next section, we study the impulse responses to some synthetic shocks defined by specific term structures. This is just one application. Many others are possible. For example, synthetic news shocks could be useful in applying the method of McKay and Wolf (2023) to generate counterfactual policy responses robust to the Lucas critique. Another possibility is to create synthetic shocks which match the term structure of a target EMPS. This would allow the construction of a comparable shock series over periods where the target series is unavailable, e.g. approximately extending to more recent data. Moreover, the broader principle is not limited just to monetary shocks. In general, one could apply our methods to other sources of macro fluctuations with partially anticipated components, such as fiscal or technological shocks.

5.2 Synthetic Surprise and News

To estimate synthetic MPS, we take a step to improve parsimony. Many of the EMPS are estimated in a similar way, and have relatively colinear term structures; Figure 3 presents their absolute correlations. Therefore, we selected a subset of five EMPS that are relatively dissimilar, as measured by the average Euclidean distance to the other vectors $\vec{\gamma}^j$. The EMPS we use for the synthetic exercise are: Aruoba-Drechsel, Miranda-Agrippino and Ricco, and the Swanson FFR, FG, and LSAP shocks. These shocks are also orthogonalized in some way to be purged of information effects, which makes the results simpler to interpret. In Appendix E, we repeat this exercise, constructed from alternative subsets.

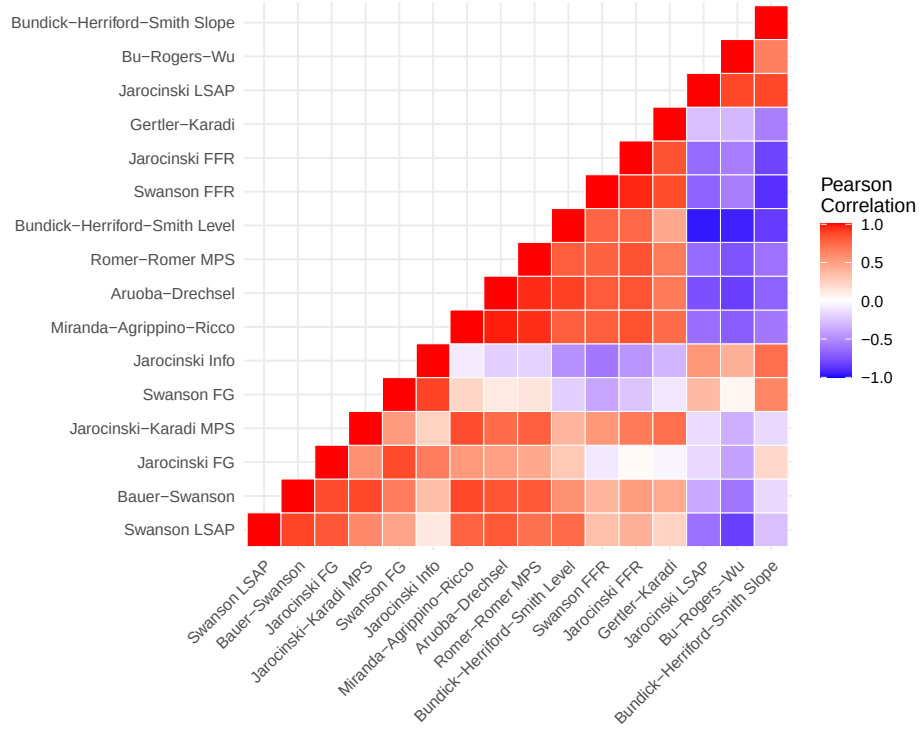


Figure 3: Term Structure Correlations

Figure shows absolute cross-correlations of the estimated term structures of candidate EMPS, ordered from least to most dissimilar top to bottom.

Using the five empirical shocks, we construct three synthetic MPS: an immediate interest rate tightening, short-term forward guidance, and long-term forward guidance. Each synthetic MPS is targeted to be an equally-weighted collection of news shocks at similar horizons. Since so many shocks come from high-frequency methods and because the relevant policy announcements can range from the first to the last day of the month, we define the immediate shock as news about the current month and 1 month ahead. The short-term

forward guidance shock contains news in the 2-6-month-ahead window and the long-term forward guidance shock contains news about the remaining year and a half.

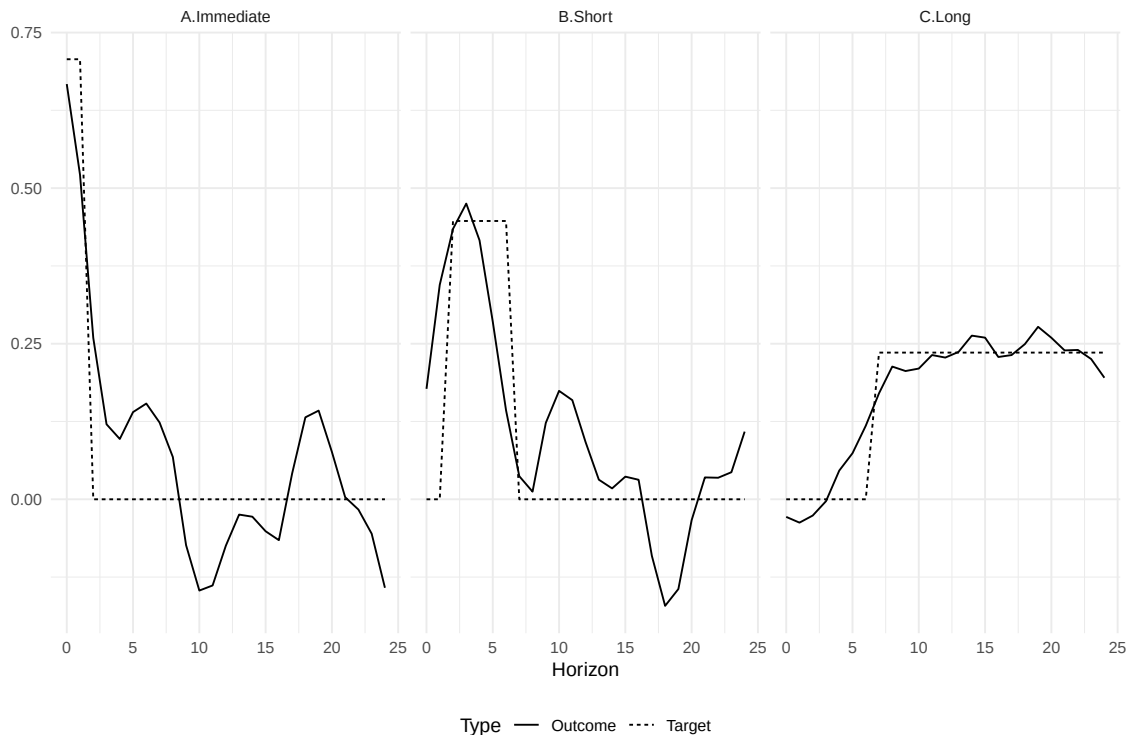


Figure 4: Target and Matched Synthetic MPS Term Structures

Figure shows target and matched term structures for synthetic policy shocks, $\tilde{\gamma}^s$ and $\tilde{\gamma}^i$ respectively.

Figure 4 plots the term structures for these synthetic MPS. The dotted lines are the target term structures, i.e. vector $\tilde{\gamma}^i$ in the notation of Proposition 7. Because we use only five empirical shocks, we cannot match these targets exactly. But five shocks gets us surprisingly close: the solid lines in Figure 4 are the actual term structures of our synthetic MPS, which approximately match the targets. A solid line corresponds to the vector $\tilde{\gamma}^s$ in Proposition 7.¹⁹ In Appendix E we use additional shocks to get closer to the target structures; doing so does not substantially change the estimated IRFs.

We estimate the effects of the synthetic MPS on the macroeconomy using a proxy VAR. Specifically, we estimate a standard VAR similar to Gertler and Karadi (2015), which includes 1-year Treasury yields, log CPI, log industrial production, and the excess bond

¹⁹Note that the target term structures combine news over multiple horizons; this contrasts to the single horizon examples described in Section 5.1. We found that the empirical MPS are much worse at accurately approximating single-horizon news (i.e. standard basis vectors) than the multiple-horizon targets that we adopted.

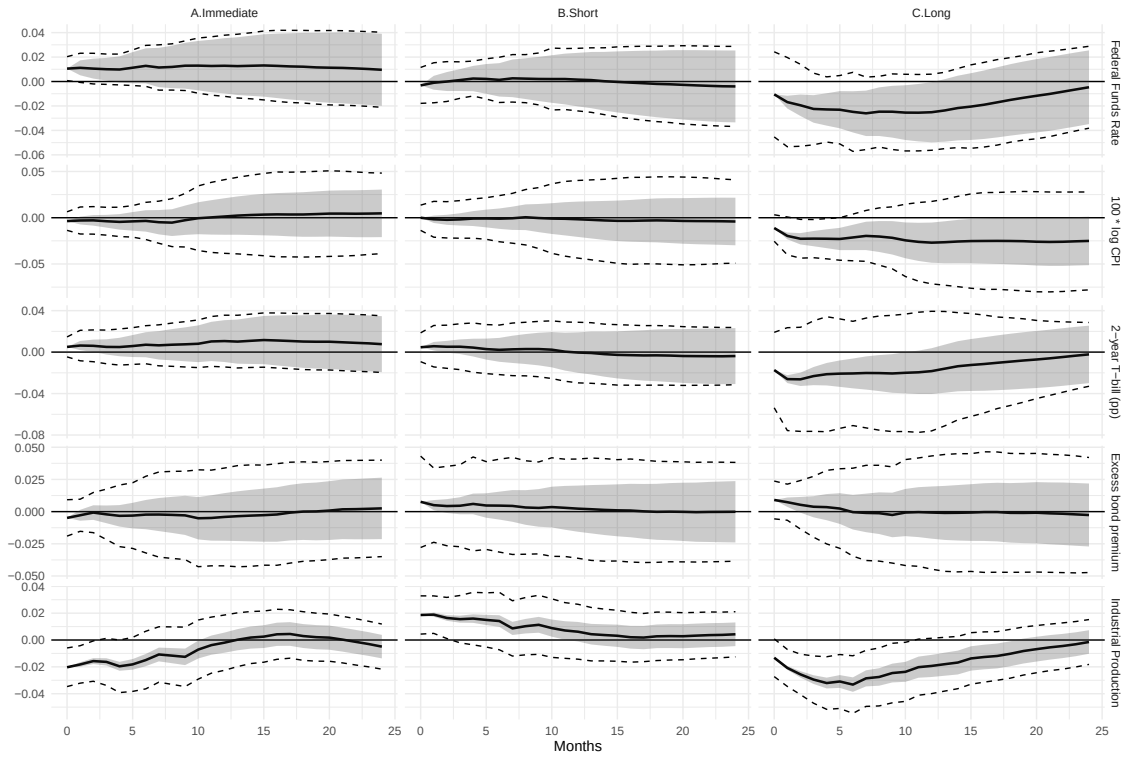


Figure 5: Impulse Responses to Synthetic Shocks

Figure shows impulse responses from a VAR to synthetic monetary policy shocks, whose term structures of monetary policy news appear in Figure 4. The first column shows responses to the immediate shock, which has news on impact and in the first following month. The second column shows responses to the short-run news shock, which has news about months 2 – 6. The third column shows responses to the long-run news shock, which has news about months 7 – 24. The shaded region indicates the 90% confidence interval conditional on the impact values; the dotted lines indicate the 90 % confidence intervals accounting for impact uncertainty. The VAR lag length is 11 and is chosen by AIC.

premium (EBP). We pick these variables so that our VAR is as close as possible to the “standard” framework in the literature. One cost of this is that we cannot include the exact variables that are used in our Taylor rule estimation. We then construct the responses to synthetic shocks by projecting each synthetic shock onto the reduced form residuals. This gives us the appropriate weighting on the reduced form shocks consistent with a synthetic term structure shock. An advantage of this approach is that we can apply the same method to the EMPS shock series, projecting them onto a common reduced-form VAR. This allows us to compare the impact of the shocks alone, holding the VAR autocovariances, and hence the dynamics of macroeconomic propagation, fixed.

Figure 5 presents the estimated IRFs to the immediate, short-term, and long-term synthetic MPS. The immediate shock is a clear monetary policy tightening: interest rates rise immediately and cause a contraction. There is perhaps a deflationary effect, but the

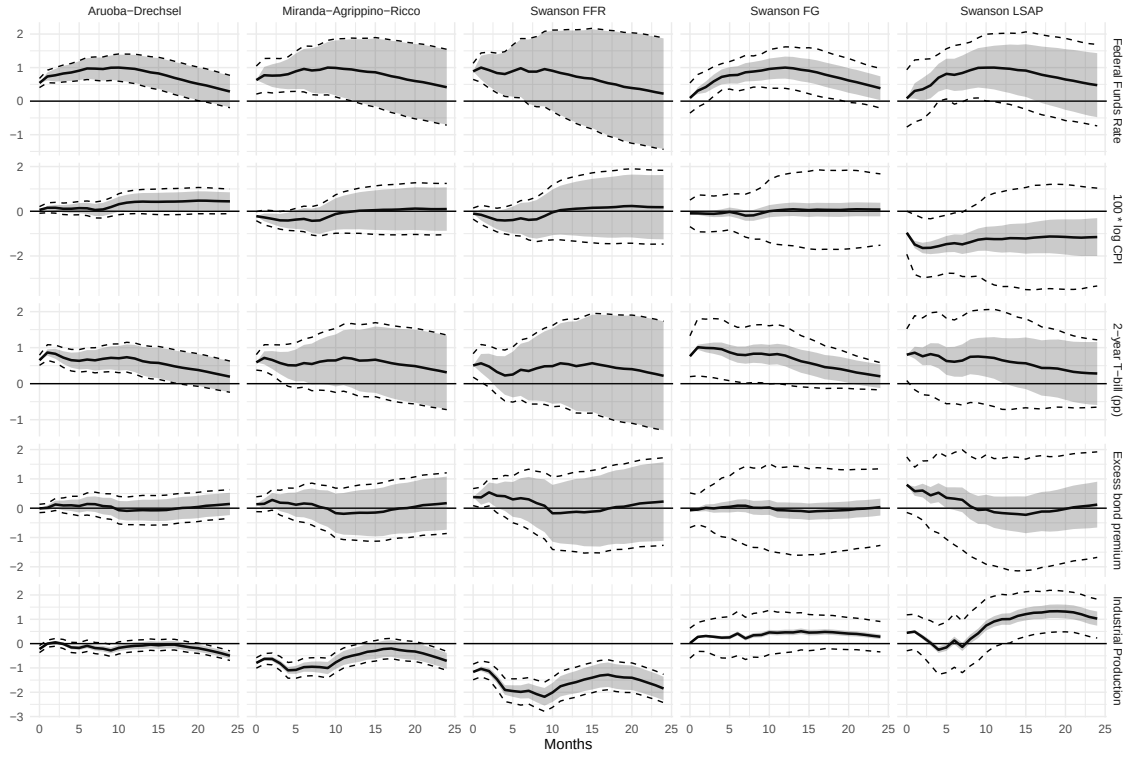


Figure 6: Impulse Responses to Empirical MPS

Figure shows impulse responses from a VAR to each of the underlying empirical MPS. Each column shows the response to a different shock. The shaded region indicates the 90% confidence interval conditional on the impact values; the dotted lines indicate the 90 % confidence intervals accounting for impact uncertainty. The VAR lag length is 11 and is chosen by AIC.

price level response is not statistically different from zero. These effects are consistent with the two most important components of the synthetic surprise: the Swanson FFR and Aruoba-Drechsel shocks are the closest EMPS to true policy surprises, and both cause large declines in real activity and near-zero price effects.

The short-run news shock has no effect on rates, and is expansionary in the initial year. Indeed, the expansionary effect is what allows rates to remain unchanged when the monetary policy residual increases. Real activity rises on impact, then falls. When the news shock manifests and policy residuals increase, real activity is declining, so the endogenous policy response is to loosen, offsetting the exogenous tightening. And altogether, the short run shock has no effect on prices at any horizon. In contrast, the long-run news shock is the only deflationary one, matching the textbook New Keynesian prediction. Counterintuitively it decreases rates in the short run, but this is also qualitatively consistent with the textbook model: the long-run forward guidance shock is persistently contractionary, so short-term monetary policy responds endogenously through the Taylor rule.

These results reveal that empirical MPS contain heterogeneous effects from news at different horizons, but this heterogeneity is unobserved without breaking apart the term structure into its components. To illustrate this finding, Figure 6 plots the IRFs for each component EMPS, as estimated by the same standard VAR. The empirical MPS are relatively homogeneous: they all predict higher rates, reduced real activity (except for the FG and LSAP shocks which do not move the FFR), and zero or modest deflation. So how can the synthetic shock IRFs look so different from the component IRFs? The key is that even though the component IRFs look relatively similar they are linearly independent, so they span a wide variety of potential outcomes. Thus different linear combinations of EMPS can produce IRFs that are strikingly different from those plotted in Figure 6 .

We learn that the relatively homogeneous empirical MPS are mixing heterogeneous effects of different news horizons. The standard rate increases are driven by *immediate* and *short-run news*. The standard deflation is driven almost entirely by *long-run news*. And the rapid contractions are not a property of forward guidance, but instead due to *immediate news*. How are we able to uncover these lessons? Because even though they are qualitatively similar, quantitative variation in the EMPS impulse responses are associated with variation in their term structures of monetary policy news.

Some of these results are consistent with the standard New Keynesian model, while some are puzzling. For example, it is standard that a surprise policy tightening is contractive, and it is typical that future monetary policy contractions can cause rates to decline in anticipation. However, the short-run expansionary effect of a short-run news shock is hard to explain with the textbook model, given that this shock also raises rates. Historically this type of “output puzzle” has been hypothesized to be caused by central bank information effects, i.e. the Fed’s actions reveal its private information about the economy. However, the EMPS that we employ are all orthogonalized to some degree, and yet still feature an output puzzle from the short-run synthetic shock. Moreover, when we repeat our exercise using alternative smaller and larger sets of component shocks (Appendix E), the results still resemble our baseline.

More substantially, it is hard to reconcile the sharp contraction of the immediate policy shock with a New Keynesian explanation when there is effectively zero effect on prices. This tension is well known (Ramey, 2016), but many shocks from the recent generation either suffered from the historical “price puzzles” (as in Aruoba and Drechsel (2024)) or caused at most a small price decline (as in Miranda-Agrippino and Ricco (2021)). Our analysis does not resolve this tension, but it does offer a richer characterization of it. Surprises do not cause large deflation; rather, the largest deflationary effects of EMPS are driven by the long-run forward guidance components.

6 Robustness

In this section, we describe how our results depend on several assumptions made in our approach.

First, we adopted many alternative approaches for our IV estimation of the Taylor rule. We discuss these in depth in Section 4.2.1, and include further robustness checks in Appendix D. We found relatively robust estimates of the Taylor rule, particularly the inflation coefficient. But how sensitive are our term structure estimates to these assumptions?

To answer this question (and others that follow) we re-estimated our main term structure summary under several alternative specifications. Figure 7 reports how the estimated term structure for each EMPS depends on the variables included in the Taylor Rule, x_t . Each panel is associated with a single EMPS, and each row in the panel is a different specification. Within each row, the bars add up to one, and each bar represents the share of the term structure that is due to news at each horizon: impact, short, medium, and long. These are the same statistics as are reported in Table 5. This figure shows that the estimated term structures are relatively consistent across specifications. For EMPS whose news is concentrated at low horizons in our baseline estimation, this also tends to be true for other specifications. For example, the Aruoba-Drechsel shock is mostly short-horizon news for all specifications, so too are the Romer & Romer narrative shocks, as well as the Swanson and Jarociński HFI Fed Funds shocks. In contrast, the Bu-Rogers-Wu shock is mostly long horizon news for all specifications, as are the Swanson and the Jarociński Forward Guidance shock.

In Figure 8, we repeat this exercise, varying the sample and estimation methods. One obvious concern about our results is that the smoothing process we apply to the term structure is driving our finding that EMPS typically have long-term effects. After all, smoothing dampens high-frequency fluctuations and so could downplay the impact of the EMPS at short horizons, risking a spurious finding that EMPS effects are at longer horizons than they actually are. Figure 8 shows that this is not the case. There, we report results for two versions, titled “low smoothing” and “no smoothing”, which respectively set the smoothing parameter to $\lambda = 10$ (approximately the lowest value across the baseline estimates for the different EMPS) and $\lambda = 0$. For even small values, the smoothing parameter does not meaningfully change our results, although when smoothing is eliminated entirely, several shocks lose most of their immediate news content. So, if anything, term structure smoothing is reallocating effects from longer horizons to shorter, thus making EMPS seem more surprise-like, not less. Changes in the instrument lags have little effect on our results, but changes in sample period do. In particular, including periods where the Federal Reserve was constrained by the zero lower bound on interest rates yields quite different results. This

is likely less a product of a true change in the term structure of monetary policy shocks, and rather a product of the Taylor rule breaking down in this period – when the ZLB binds, the Fed no longer responds to marginal changes in inflation or output.

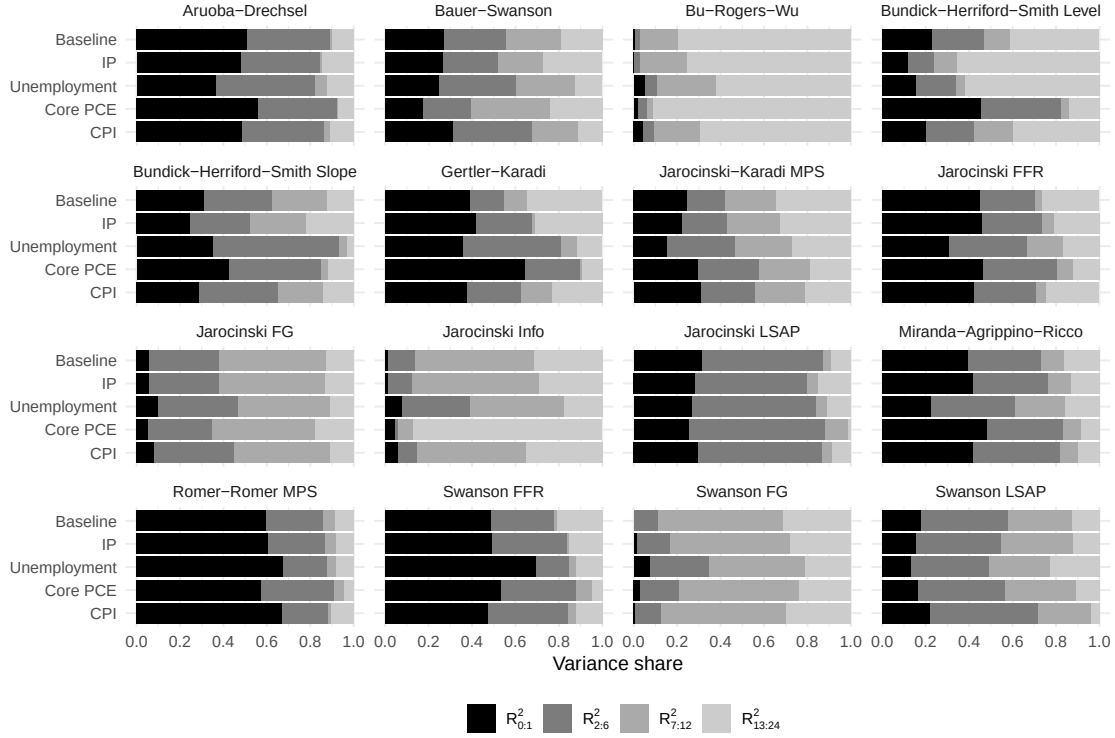


Figure 7: Term Structure Variance Decomposition: Robustness to Taylor Rule Variables.

Figure shows the how the variance decomposition changes for different versions of the estimated term structure. “Baseline” corresponds to the numbers in Table 5. Different versions (labeled on the vertical axis) correspond to the alternate Taylor rule estimation methods with the same names as in Tables 3 to 7.

7 Conclusions

In this paper, we address two important questions about the identification of monetary policy shocks.

The first is: how should we compare different estimated monetary policy shocks? The framework we develop in this paper is based on the idea that identified monetary policy shocks identify a common type of exogenous disturbance, but vary in its anticipated timing. By applying this method to identify the differences in sixteen well-known monetary shock series, we decompose each into its surprise and news components, the latter at multiple horizons. We find that most of these shocks have large news components. However, shocks

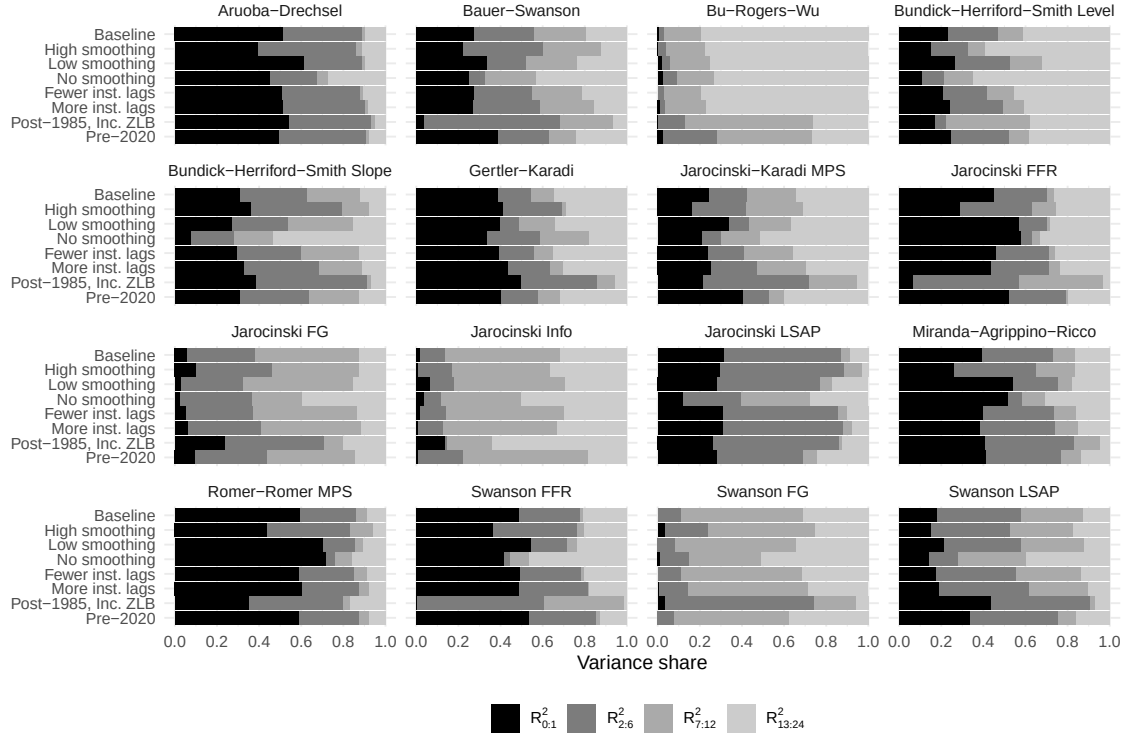


Figure 8: Term Structure Variance Decomposition: Robustness to Sample and Estimation Method

Figure shows the how the variance decomposition changes for different versions of the estimated term structure. “Baseline” corresponds to the numbers in Table 5. Different versions (labeled on the x axis) correspond to the alternate Taylor rule estimation methods with the same names as in Tables 3 to 7.

explicitly designed to capture surprises generally have shorter term structures and those aimed at forward guidance shocks longer.

Second, how can we map between empirical shocks and theory? By projecting fixed h -period ahead impulses onto imperfectly correlated empirical shocks, we can construct the responses to news shocks at multiple horizons as a linear combination of estimated impulse responses. In doing so, we are able to characterize the empirical responses of shocks which comport with theory. We show that positive monetary surprises are contractionary and deflationary, but that news at longer horizons increases output, employment, and prices. At very long horizons the effect of monetary policy news is negligible.

These results suggest several directions for future research. Most obviously, they provide a framework for evaluating future monetary policy shocks, allowing them to be compared to those already in the literature. Our synthetic shock approach also offers opportunities for many further potential applications. And our specific findings give some guidance on

how empirical identification of MPS might most valuably proceed. In particular, our results show that there is still much to be done to systematically capture monetary policy surprises distinct from news about the future. Beyond this, our findings on the measured effect of news shocks at multiple horizons set an empirical benchmark for future models to reflect.

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A Proofs

Proposition 1.

First, the 2SLS estimator for the Taylor rule is

$$\hat{\phi} = (X'P_ZX)^{-1}X'P_Zy$$

The OLS estimator for the third stage regression (9) is

$$\hat{\varrho} = (\hat{\mathbf{R}}'\hat{\mathbf{R}})^{-1}\hat{\mathbf{R}}'\hat{R}$$

where $\hat{R} = y - X\hat{\phi}$. $\hat{\mathbf{r}}_t = (y_{t-1} - x_{t-1}\hat{\phi} \quad \dots \quad y_{t-L} - x_{t-L}\hat{\phi})$ is a row vector of lagged residuals, so the matrix $\hat{\mathbf{R}}$ is constructed by

$$\hat{\mathbf{R}} = \mathbf{Y} - \mathbf{X}(I_{n_x} \otimes \hat{\phi})$$

Therefore $\hat{\nu}$ is given by

$$\begin{aligned} \hat{\nu} &= \hat{R} - \hat{\mathbf{R}}\hat{\varrho} \\ &= (y - X\hat{\phi}) - \hat{\mathbf{R}}(\hat{\mathbf{R}}'\hat{\mathbf{R}})^{-1}\hat{\mathbf{R}}'(y - X\hat{\phi}) = M_{\hat{\mathbf{R}}}M_{IV}y \end{aligned}$$

using $M_{\hat{\mathbf{R}}} = I - \hat{\mathbf{R}}(\hat{\mathbf{R}}'\hat{\mathbf{R}})^{-1}\hat{\mathbf{R}}'$ and $M_{IV} = I - X(X'P_ZX)^{-1}X'P_Z$.

The OLS estimator for the fourth stage regression (10) is

$$\hat{\gamma} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'\hat{\nu} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'M_{\hat{\mathbf{R}}}M_{IV}y$$

■

Proof of Proposition 2.

Write the estimator as

$$\begin{aligned} \hat{\gamma} &= (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'M_{\hat{\mathbf{R}}}(y - X\hat{\phi}) = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'M_{\hat{\mathbf{R}}}(R + X(\phi - \hat{\phi})) \\ &= (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'(M_{\mathbf{R}}R + (M_{\hat{\mathbf{R}}} - M_{\mathbf{R}})R + M_{\hat{\mathbf{R}}}X(\phi - \hat{\phi})) \end{aligned}$$

where $M_{\mathbf{R}} = I - \mathbf{R}(\mathbf{R}'\mathbf{R})^{-1}\mathbf{R}'$, i.e. the corresponding matrix with the true lags of Taylor residuals. $R = \mathbf{R}\varrho + \nu$ implies $M_{\mathbf{R}}R = \nu$ because ν is orthogonal to lagged residuals $\mathbf{R}$. Then $\nu = W\gamma + u$ implies

$$\hat{\gamma} = \gamma + (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'(u + (M_{\hat{\mathbf{R}}} - M_{\mathbf{R}})R + M_{\hat{\mathbf{R}}}X(\phi - \hat{\phi})) \quad (12)$$

The final three terms in equation (12) are zero in the limit. By definition $\mathbb{E}[\mathbf{w}_t u_t] = 0$ so

$\text{plim}((\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'u) = 0$. By definition $\mathbb{E}[\mathbf{r}_t\nu_t] = 0$ so $\text{plim}(M_{\hat{\mathbf{R}}}) = M_{\mathbf{R}}$. By assumption $\mathbb{E}[z_t r_t] = 0$ so $\text{plim}(\hat{\phi}) = \phi$. Therefore equation (12) implies

$$\text{plim}(\hat{\gamma}) = \gamma$$

■

Proof of Proposition 4. By equation (5), the variance of the expectation of ν_t conditional on w_{t-k}^j is

$$\begin{aligned} \text{Var}(\mathbb{E}[\nu_t|w_{t-k}^j]) &= \text{Var}(\mathbb{E}[\sum_{h=0}^{H_w} \gamma_h^j w_{t-h}^j + u_t^j | w_{t-k}^j]) \\ &= \text{Var}(\mathbb{E}[\gamma_h^j w_{t-k}^j | w_{t-k}^j]) = \text{Var}(\gamma_h^j w_{t-k}^j) = (\gamma_k^j)^2 \text{Var}(w_t^j) \end{aligned}$$

because the past EMPS is homoskedastic white noise, and orthogonal to w_t^j . Similarly, the total variance conditional on the history of EMPS is

$$\begin{aligned} \text{Var}(\nu_t | \{w_{t-h}^j\}_{h=0}^{H_w}) &= \text{Var}(\mathbb{E}[\sum_{h=0}^{H_w} \gamma_h^j w_{t-h}^j + u_t^j | \{w_{t-k}^j\}_{k=0}^{H_w}]) \\ &= \text{Var}(\sum_{h=0}^{H_w} \gamma_h^j w_{t-h}^j) = \sum_{h=0}^{H_w} \text{Var}(\gamma_h^j w_{t-h}^j) = \left(\sum_{h=0}^{H_w} (\gamma_h^j)^2 \right) \text{Var}(w_t^j) \end{aligned}$$

Combining these two equations gives the ratio

$$\frac{\text{Var}(\nu_t | w_{t-k}^j)}{\text{Var}(\nu_t | \{w_{t-h}^j\}_{h=0}^{H_w})} = \frac{(\gamma_k^j)^2}{\sum_{h=0}^{H_w} (\gamma_h^j)^2}$$

■

Proof of Proposition 5. By equation (4), the EMPS w_t^c can be written as

$$\begin{aligned} w_t^c &= \lambda_a w_t^a + \lambda_b w_t^b \\ &= \lambda_a \sum_{h=0}^{H_{w^a}} \beta_h^a \nu_{h,t} + \lambda_a \xi_t^a + \lambda_b \sum_{h=0}^{H_{w^b}} \beta_h^b \nu_{h,t} + \lambda_b \xi_t^b \\ &= \sum_{h=0}^{H_{w^c}} \beta_h^c \nu_{h,t} + \xi_t^c \end{aligned}$$

where $\beta_h^c = \lambda_a \beta_h^a + \lambda_b \beta_h^b$, $H_{w^c} = \max\{H_{w^a}, H_{w^b}\}$ and $\xi_t^c = \lambda_a \xi_t^a + \lambda_b \xi_t^b$ is orthogonal to $\nu_{h,t}$

for all h . By equation (6), the term structure coefficients are given by

$$\begin{aligned}\gamma_h^c &= \left(\lambda_a \beta_h^a + \lambda_b \beta_h^b \right) \frac{Var(\nu_{h,t})}{Var(w_t^c)} \\ &= \lambda_a \gamma_h^a \frac{Var(w_t^a)}{Var(w_t^c)} + \lambda_b \gamma_h^b \frac{Var(w_t^b)}{Var(w_t^c)}\end{aligned}$$

When $Var(w_t^a)$ and $Var(w_t^b)$ are normalized to 1, the vector form of this equation is

$$\begin{aligned}\vec{\gamma}^c &= \lambda_a \vec{\gamma}^a \frac{1}{Var(w_t^c)} + \lambda_b \vec{\gamma}^b \frac{1}{Var(w_t^c)} \\ &\propto \lambda_a \vec{\gamma}^a + \lambda_b \vec{\gamma}^b\end{aligned}$$

■

Proof of Proposition 7. The synthetic MPS $\vec{\gamma}^s$ must be in the span of the observed EMPS term structures, i.e. the columns of $\Gamma_{\mathcal{J}}$. The vector in this span minimizing $\|\vec{\gamma}^i - \vec{\gamma}\|_2$ is the projection of $\vec{\gamma}^i$ onto the span of the columns of $\Gamma_{\mathcal{J}}$. This is given by the familiar expression

$$\vec{\gamma}^s = \Gamma_{\mathcal{J}} (\Gamma'_{\mathcal{J}} \Gamma_{\mathcal{J}})^{-1} \Gamma'_{\mathcal{J}} \vec{\gamma}^i$$

■

B Smooth Term Structures

This appendix describes how to estimate the smoothed term structures, analogous to the smooth local projections of Barnichon and Brownlees (2019). First we derive how to estimate smooth local projections in closed form. In particular, we show that there is a shortcut such that transformation with B-splines is not needed at all; the local projection can be estimated by ridge regression using a suitable penalty matrix. Then we show how to apply the smoothing in the context of our broader method.

B.1 Smooth Local Projections

Consider the following local projection for $h = 0, 1, \dots, H$:

$$y_{t+h} = w_t \gamma_h + \epsilon_{h,t+h}$$

where y_{t+h} is the outcome variable of interest, w_t is an exogenous shock, and $\epsilon_{h,t+h}$ is the error term. If w_t is white noise, then the local projection coefficients can be estimated from the following regression:

$$y_t = \sum_{h=0}^H \gamma_h w_{t-h} + \epsilon_t \quad (13)$$

The smooth local projection approach is to approximate the γ_h coefficients with B-splines, which are indexed piecewise polynomial functions $B_0(h), B_1(h), \dots, B_K(h)$. The coefficients are given by

$$\gamma_h = \sum_0^K \alpha_k B_k(h)$$

where α_k are coefficients to be estimated. We can rewrite the local projection regression as

$$\begin{aligned} y_t &= \sum_{h=0}^H \sum_0^K \alpha_k B_k(h) w_{t-h} + \epsilon_t \\ &= \sum_0^K \alpha_k v_{t-h} + \epsilon_t \end{aligned}$$

where

$$v_{t-h} = \sum_{h=0}^H B_k(h) w_{t-h} \quad (14)$$

is a *smoothed* version of the shock. The coefficients α_k can be estimated by OLS.

A vector representation is useful. Let $\vec{w}_t$ be the $H+1$ -dimensional row vector of shocks at time t , and $\vec{v}_t$ be the $H+1$ -dimensional row vector of smoothed shocks at time t . They

are related by

$$\vec{v}_t = \vec{w}_t B$$

where B is the $(H+1) \times (H+1)$ matrix of B-spline basis functions, sampled at appropriate points to recover equation (14). Stack the vectors into matrices, so that V is the $T \times (H+1)$ matrix of smoothed shock vectors, W is the $T \times (H+1)$ matrix of shock vectors, and y is the $T \times 1$ vector of outcomes. The smooth local projection regression is written

$$y = V\alpha + \epsilon$$

where α is the $K+1$ -dimensional vector of coefficients, and ϵ is the $T \times 1$ vector of errors. The coefficients from the original form $Y = W\gamma + \epsilon$ can be recovered by

$$\gamma = B\alpha$$

because $WB = V$.

Barnichon and Brownlees (2019) estimate the smooth local projections by ridge regression. An appropriate penalty term gives the interpretation that the local projection is shrunk towards a lower order polynomial. The ridge regression estimator is

$$\begin{aligned} \hat{\alpha}_\lambda &= \arg \min_{\alpha} (y - V\alpha)' (y - V\alpha) + \lambda \alpha' \mathbf{P} \alpha \\ &= (V'V + \lambda \mathbf{P})^{-1} V'Y \end{aligned}$$

where λ is a positive shrinkage parameter, and $\mathbf{P}$ is the penalty matrix. λ can be chosen by cross-validation. For the canonical smooth local projections the penalty matrix is

$$\mathbf{P} = \mathbf{D}_r' \mathbf{D}_r$$

where $\mathbf{D}_r$ is the r th difference matrix.

Because the estimated original coefficients are related by $\hat{\gamma}_\lambda = B\hat{\alpha}_\lambda$, there is a short-cut to smooth local projections that skips the transformation step entirely:

Proposition 8 *The smooth local projection coefficient vector $\hat{\gamma}_\lambda$ can be found by estimating equation (13) by ridge regression with penalty matrix*

$$\mathbf{P}_B = (B^{-1})' \mathbf{P} B^{-1}$$

so that the estimate is given by

$$\hat{\gamma}_\lambda = (W'W + \lambda \mathbf{P}_B)^{-1} W'y$$

Proof. The relationship $\hat{\gamma}_\lambda = B\hat{\alpha}_\lambda$ and the expression for the ridge regression estimator $\hat{\alpha}_\lambda$ imply

$$\begin{aligned}\hat{\gamma}_\lambda &= B(V'V + \lambda\mathbf{P})^{-1}V'y \\ &= B(B'W'WB + \lambda\mathbf{P})^{-1}B'W'y \\ &= (W'W + \lambda(B^{-1})'\mathbf{P}B^{-1})^{-1}W'y\end{aligned}$$

The definition $\mathbf{P}_B = (B^{-1})'\mathbf{P}B^{-1}$ gives the proposed expression, which is equivalent to the ridge regression estimator with penalty matrix $\mathbf{P}_B$. ■

Ridge regression also has closed form standard errors. The conditional variance of the ridge regressor is

$$Var(\hat{\alpha}|V) = \sigma^2 (V'V + \lambda\mathbf{P})^{-1}V'V(V'V + \lambda\mathbf{P})^{-1}$$

where σ^2 is the error variance. Returning to the original coefficients, the conditional variance is

$$\begin{aligned}Var(\hat{\gamma}_\lambda|V) &= \sigma^2 B(V'V + \lambda\mathbf{P})^{-1}V'V(V'V + \lambda\mathbf{P})^{-1}B' \\ Var(\hat{\gamma}_\lambda|W) &= \sigma^2 B(B'W'WB + \lambda\mathbf{P})^{-1}B'W'WB(B'W'WB + \lambda\mathbf{P})^{-1}B' \\ &= \sigma^2 (W'W + \lambda(B^{-1})'\mathbf{P}B^{-1})^{-1}W'W(W'W + \lambda(B^{-1})'\mathbf{P}B^{-1})^{-1}\end{aligned}$$

B.2 Smoothed Term Structures

We can apply the smooth local projection method to the term structure estimation. The final step of the four-stage procedure is to regress the estimated policy residuals $\hat{\nu}_t$ onto lags of the EMPS w_t^j . The smooth local projection method is directly applicable to equation (5). The regression is

$$\begin{aligned}\nu_t &= \sum_{h=0}^{H_w} \gamma_h^j w_{t-h}^j + u_t^j \\ &= \sum_{h=0}^{H_w} \sum_0^K \alpha_k^j B_k(h) w_{t-h}^j + u_t^j \\ &= \sum_0^K \alpha_k^j v_{t-h}^j + u_t^j\end{aligned}$$

where again v_{t-h}^j is the smoothed shock.

The ridge regression estimator for the vector of α_k^j coefficients is

$$\hat{\alpha}_\lambda = (V'V + \lambda\mathbf{P})^{-1}V'\hat{\nu}$$

where V is the vector of transformed shocks v_t^j . The penalty matrix $\lambda \mathbf{P}$ is the same as in the previous section. The coefficients are related to the term structure coefficients by

$$\hat{\gamma}_\lambda = B \hat{\alpha}_\lambda$$

and the vector $\hat{\nu}$ is given in matrix notation by $\hat{\nu} = M_{\hat{\mathbf{R}}} M_{IV} y$, so the smoothed term structure estimator is

$$\hat{\gamma}_\lambda = B (V'V + \lambda \mathbf{P})^{-1} V' M_{\hat{\mathbf{R}}} M_{IV} y$$

In terms of untransformed shocks, the estimator is

$$\hat{\gamma}_\lambda = (W'W + \lambda \mathbf{P}_B)^{-1} W' M_{\hat{\mathbf{R}}} M_{IV} y$$

The penalized regression of course is biased, with probability limit

$$\hat{\gamma}_\lambda \xrightarrow{p} (W'W + \lambda \mathbf{P}_B)^{-1} W'W \gamma$$

following the same logic as in the proof of Proposition 2.

C The Term Structures of Interest Rates Versus Monetary Policy News

A frequently-asked question we receive is: Why is the term structure of news not easily identified from the term structure of interest rates? EMPS have different effects on interest rates at different maturities; we have made a theoretical case why simply estimating these effects will not recover the news structure. In this appendix, we illustrate the case quantitatively, and demonstrate that the differences are empirically relevant.²⁰

Examining how EMPS affects the term structure of interest rates appears to be a reasonable idea for estimating the news content of an EMPS. It is intuitive: long-term interest rates depend on expectations about future policy rates, and the news encoded in an EMPS also affects future policy rates. But simply estimating how an EMPS affects the path of future policy rates will fail to recover the term structure of monetary policy news because future rates do not only depend on future policy residuals. Future policy rates also depend on the endogenous policy response to macroeconomic conditions. Indeed, the endogenous response explains most of the variation in policy rates. So directly regressing future policy rates on an EMPS will conflate the exogenous policy news with the endogenous response to whatever effects the EMPS has on inflation, unemployment, and so forth. This is why

²⁰We are grateful to Michael Bauer for suggesting this exercise.

our 4-stage method requires an IV step to estimate the policy rules in order to account for endogenous responses.

Do the endogenous responses matter quantitatively? Yes. To demonstrate, we estimate an IRF by local projection for how each EMPS affects future federal funds rates. To be directly comparable, we apply the same smoothing and sample construction decisions as in our main method. Thus we are estimating our fourth stage equation (5) except with the federal funds rate on the left-hand side instead of the policy shock ν_t . We repeat this estimation for each EMPS from our main analysis and plot the estimates in Figure 9.

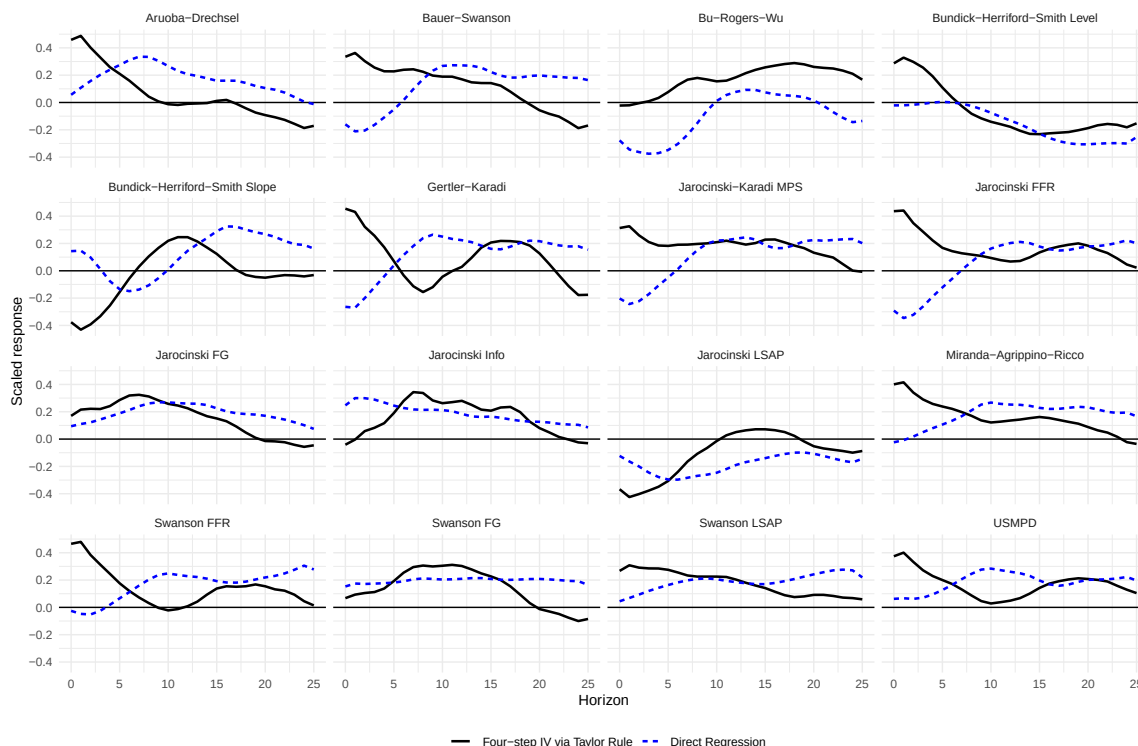


Figure 9: Estimated Term Structures

Figure shows the estimated γ_h^j coefficients, i.e. the impact of each identified monetary policy shock h periods later on two variables. The solid black line corresponds to the monetary policy innovation $\hat{\nu}_{t+h}$; the dashed blue line is from the direct regression of the Federal Funds rate on lagged shocks, which includes endogenous responses of macro variables to the measured shock. The coefficients are normalized to have a unit squared sum. Term structures are smoothed using $\lambda = 30$.

What would one expect to see in these plots if future spot rates were sufficient to reveal the term structure of monetary policy news? The black and blue curves in Figure 9 would be similar, e.g. the Aruoba-Drechsel shock, which mainly consists of short-run news, would mainly affect the FFR at short-run horizons.²¹

²¹Indeed, many of the EMPS have small effects on short-run Federal Funds rates, which is one reason that

Instead, the interest rate term structure effects are noticeably different from our estimated term structures of monetary policy news. The endogenous component of the policy rule is large, and must be accounted for to accurately measure the news content of an EMPS. The differences are quantitatively large, but also qualitatively surprising, with small or even negative impact responses for many estimated shocks.

This raises an obvious question: How can well-identified EMPS have such small or negative effects on policy rates at short horizons? Our term structure estimation helps provide an answer. Even though EMPS such as the Swanson FFR shock are largely determined by short-term policy news, they still contain substantial longer-term news. The direct IRF (the blue line in Figure 9) is the sum of how these components affect the policy rate. If longer-term news has a sufficiently large anticipation effect, where the economy slows immediately and causes an endogenous reduction of the policy rule, this could offset short-term effect, producing the patterns shown in Figure 9. Such large anticipation effects are possible in standard new Keynesian models (Carlstrom et al., 2015), and are also consistent with our estimated effects of synthetic long-term news shocks (Figure 5).

D Further Taylor Rule Specification Alternatives

Consistent estimation of the Taylor rule is crucial for our estimation exercise. In Section 4.2.1 we explored how robust our estimates are to alternative measures of inflation and real activity. In this appendix, we explore further alternative specifications.

Table 6 presents how our results depend on the sample period used. Our results are broadly robust across these choices, except in the extreme case where we both begin the sample after 1985 and include the zero-lower-bound (ZLB) period.

In our baseline IV estimation, we used three types of structural shocks as instruments, listed in Table 2. In case one of the instruments fails the exclusion restrictions, we also repeat our analysis with different subsets of instruments. Table 7 reports the estimates. The inflation coefficient in the Taylor rule is robust to these choices, although the GDP coefficient changes substantially.

Finally, we include a variety of OLS estimates for the Taylor rule, to compare our approach with the typical method in the literature. These are reported in Table 8 which reveals that OLS estimates are highly attenuated compared to our IV results. In addition to the expected bias, OLS estimates are also highly sensitive to changes in the regression specification.

1-year or 2-year yields are typically used as the policy indicator in VAR analysis. These shocks can have much larger effects on short-run policy residuals because they tend to have clear effects on inflation and real activity, which enter the endogenous component of the policy rule.

	Baseline	Inc. ZLB	Post-1985	Post-1985, Inc. ZLB	Pre-2020
12-month PCE Inflation, demeaned	1.517*** (0.097)	1.138*** (0.098)	0.453** (0.194)	0.091 (0.162)	1.501*** (0.076)
Private nonfarm payroll employment, departures from quadratic log trend, demeaned	0.370*** (0.036)	0.186*** (0.042)	0.237*** (0.039)	0.373*** (0.041)	0.372*** (0.033)
Residual autocorrelation	0.95	0.97	0.98	0.99	0.96
R^2	0.58	0.59	0.58	0.61	0.68
Observations	569	569	449	449	521

Table 6: Estimated Taylor Rule Parameters: Different samples

Table reports the estimated Taylor rule parameters from the second stage of the four-stage method using instrumental variables. Each column corresponds to a different sample period. The baseline sample runs from 1975:M1-2020:M2, excluding the ZLB period. Standard errors are reported in parentheses.

	Baseline	Fewer inst. lags	More inst. lags	$n_{H_v}=12$	Omit gov.	Omit oil	Omit other
12-month PCE Inflation, demeaned	1.517*** (0.097)	1.578*** (0.144)	1.358*** (0.077)	1.518*** (0.099)	1.652*** (0.118)	1.111*** (0.141)	1.652*** (0.118)
Private nonfarm payroll employment, departures from quadratic log trend, demeaned	0.370*** (0.036)	0.318*** (0.059)	0.359*** (0.024)	0.367*** (0.037)	0.118** (0.047)	0.514*** (0.041)	0.118** (0.047)
Residual autocorrelation	0.95	0.95	0.96	0.95	0.96	0.97	0.96
R^2	0.58	0.57	0.61	0.59	0.50	0.57	0.50
Observations	569	565	575	581	569	569	569

Table 7: Estimated Taylor Rule Parameters: Different instruments

Table reports the estimated Taylor rule parameters from the second stage of the four-stage method using instrumental variables. Residual autocorrelation is the first order autocorrelation of the monetary policy residual, r_t . R^2 is calculated as the fraction of variance in the policy rate explained by the contemporaneous systematic part of monetary policy, $x_t\phi$, and so $1 - R^2$ is that explained by the monetary policy residual, r_t . Standard errors are reported in parentheses.

	Baseline	OLS	OLS, no RHS int. rate	OLS, 1 lagged int. rate	OLS, no lags	OLS, 12 lags	OLS, 12 lags no int. rate
12-month PCE Inflation, demeaned	1.517*** (0.097)	0.038 (0.089)	0.282 (0.487)	0.147* (0.086)	-0.143 (0.483)	0.014 (0.082)	0.629 (0.431)
Private nonfarm payroll employment, departures from quadratic log trend, demeaned	0.370*** (0.036)	0.488*** (0.160)	-0.004 (0.934)	0.801*** (0.178)	1.369* (0.722)	0.523*** (0.182)	0.616 (0.816)
Residual autocorrelation	0.95	-0.01	0.95	0.29	0.95	-0.03	0.94
R^2	0.58	0.96	0.65	0.97	0.59	0.94	0.55
Observations	569	582	582	583	583	576	576

Table 8: Estimated Taylor Rule Parameters: OLS estimates

Table reports the estimated Taylor rule parameters from the second stage of the four-stage method using instrumental variables. Residual autocorrelation is the first order autocorrelation of the monetary policy residual, r_t . R^2 is calculated as the fraction of variance in the policy rate explained by the contemporaneous systematic part of monetary policy, $x_t\phi$, and so $1 - R^2$ is that explained by the monetary policy residual, r_t . Standard errors are reported in parentheses.

E Alternative Synthetic Shocks

In Section 5, we used a set of modern EMPS to construct the synthetic MPS. In this appendix, we repeat the exercise with alternative sets of EMPS. Broadly speaking, the estimated IRFs look similar to the baseline case: the immediate shock is contractionary with little effect on prices, the short-run shock is expansionary at some horizons, and the long-run shock is deflationary and reduces rates on impact.

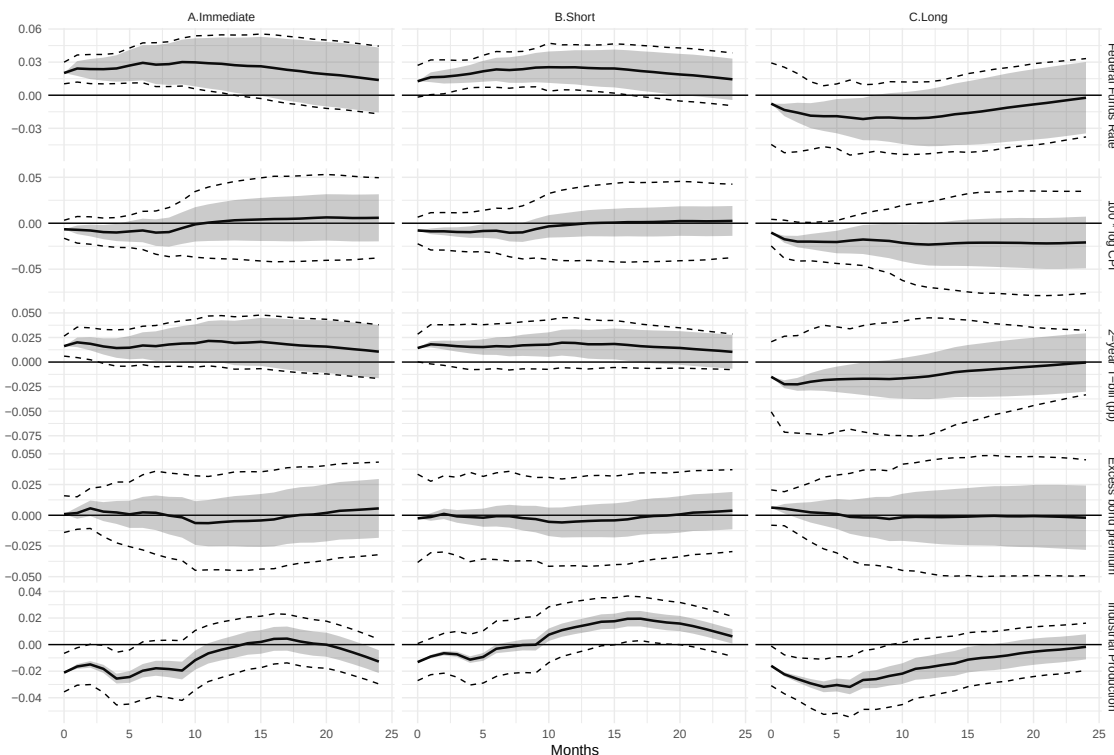


Figure 10: Impulse Responses to Synthetic Shocks (Smaller Set)

Figure shows impulse responses from a VAR to synthetic monetary policy shocks. The first column shows responses to the immediate shock, which has news on impact and in the first following month. The second column shows responses to the short-run news shock, which has news about months 2 – 6. The third column shows responses to the long-run news shock, which has news about months 7 – 24. The shaded region indicates the 90% confidence interval conditional on the impact values; the dotted lines indicate the 90 % confidence intervals accounting for impact uncertainty due to the proxy VAR. The VAR lag length is 11 and is chosen by AIC.

In our first alternative, we employ a smaller set of EMPS, restricted to a few shocks that are thought to represent traditional monetary policy tools. In this implementation, we drop the Swanson LSAP shock from the baseline, which may capture non-traditional tools used by the Fed beyond classic interest rate policy. Figure 10 plots the estimated IRFs to the synthetic shocks estimated from this smaller set. They closely resemble the baseline

approach, although now the short-run news shock does not feature a temporary production expansion.

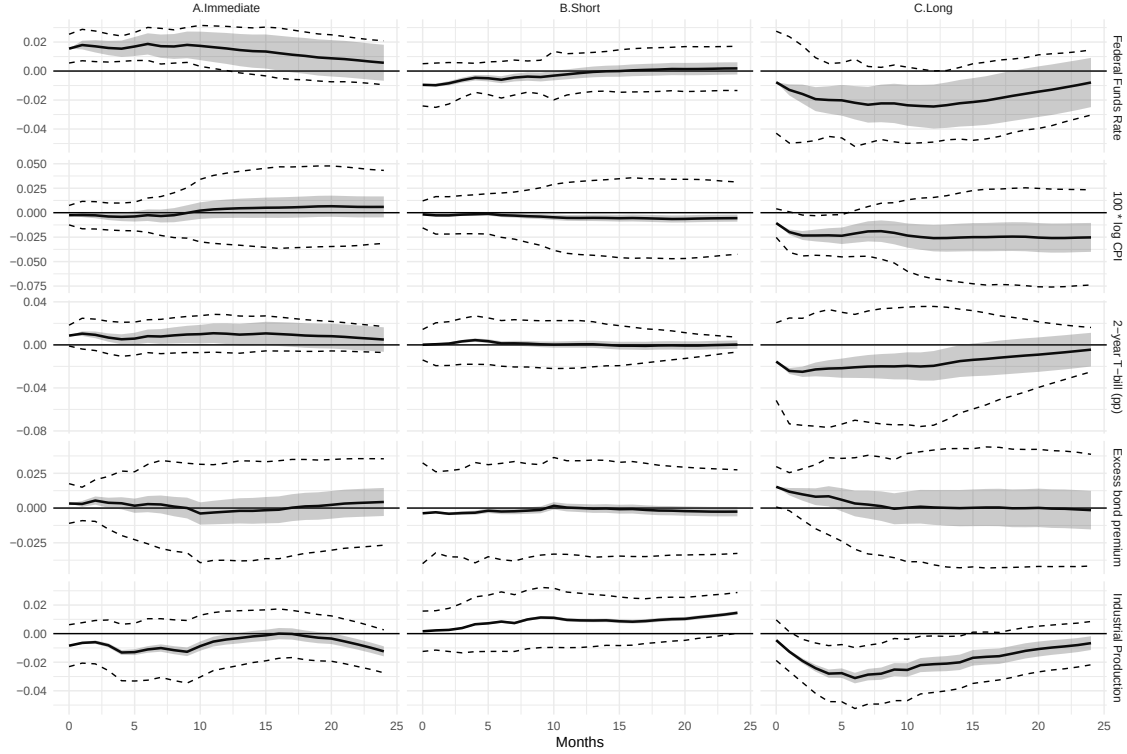


Figure 11: Impulse Responses to Synthetic Shocks (Larger Set)

Figure shows impulse responses from a VAR to synthetic monetary policy shocks. The first column shows responses to the immediate shock, which has news on impact and in the first following month. The second column shows responses to the short-run news shock, which has news about months 2 – 6. The third column shows responses to the long-run news shock, which has news about months 7 – 24. The shaded region indicates the 90% confidence interval conditional on the impact values; the dotted lines indicate the 90 % confidence intervals accounting for impact uncertainty due to the proxy VAR. The VAR lag length is 11 and is chosen by AIC.

In our second alternative, we employ an expanded set of EMPS in order to more accurately approximate the target synthetic shock structure (Figure 2). To the baseline we add the Jarociński and Karadi (2020) shock, which controls for central bank information in a different way than Miranda-Agrippino and Ricco (2021). And even though they have relatively similar term structures, the shocks are not perfectly correlated and cover different samples improving our statistical power. Figure 11 plots the IRFs. The main difference relative to the baseline is that the IRFs are more precisely estimated. Beyond this, the short-run news shock has a more negative effect on rates and a longer-lived expansion.

Our final alternative adds the Bundick and Smith (2020) term structure uncertainty level shock, which is constructed from high-frequency data to capture yet another channel

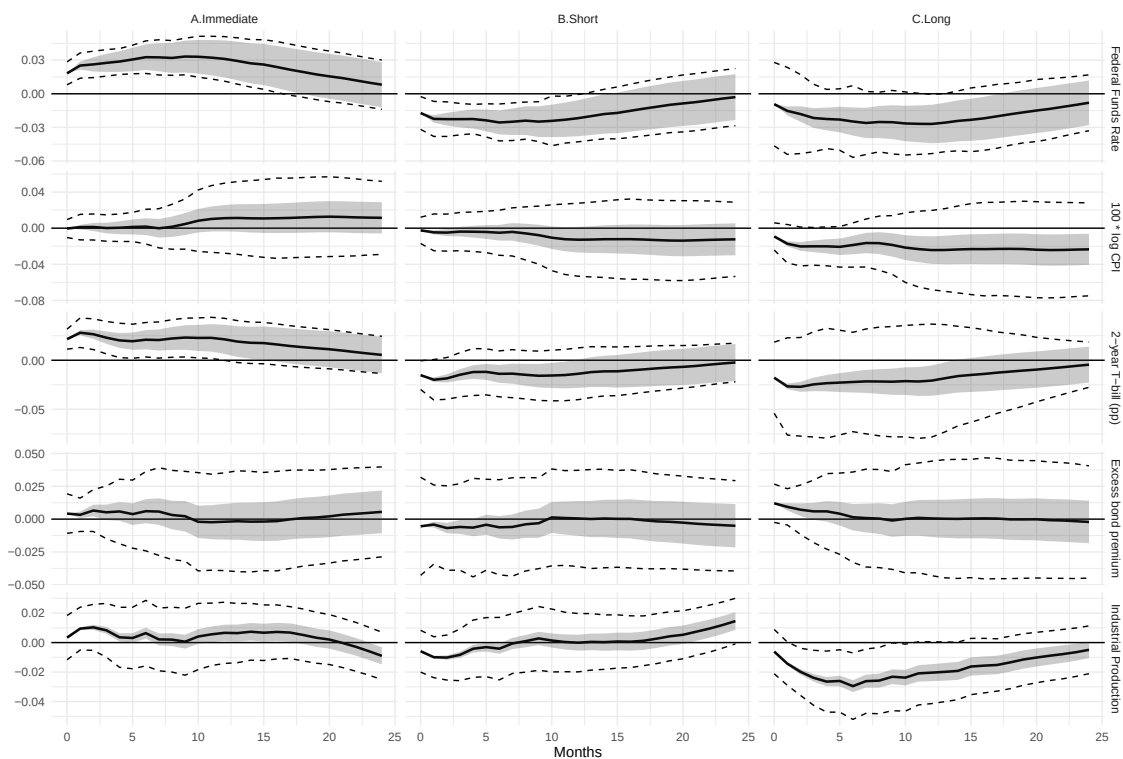


Figure 12: Impulse Responses to Synthetic Shocks (Largest Set)

Figure shows impulse responses from a VAR to synthetic monetary policy shocks. The first column shows responses to the immediate shock, which has news on impact and in the first following month. The second column shows responses to the short-run news shock, which has news about months 2 – 6. The third column shows responses to the long-run news shock, which has news about months 7 – 24. The shaded region indicates the 90% confidence interval conditional on the impact values; the dotted lines indicate the 90 % confidence intervals accounting for impact uncertainty. The VAR lag length is 11 and is chosen by AIC.

of monetary policy. Altogether, we use seven different EMPS to construct this series of synthetic shocks. The IRFs of the synthetic shocks are plotted in Figure 12, and the IRFs of each component are plotted in Figure 13. Again, the main differences between these results and the baseline specification are in the effects of short-run news. The interest rate effect is more negative, and real activity now falls for several months before a mild expansion.

F Monte Carlo Validation

To validate our method and to check some of its properties, we conduct a Monte Carlo experiment. We simulate data from the motivating three-equation New Keynesian model from Section 2, extended to include additive shocks to the Euler equation (a “demand”

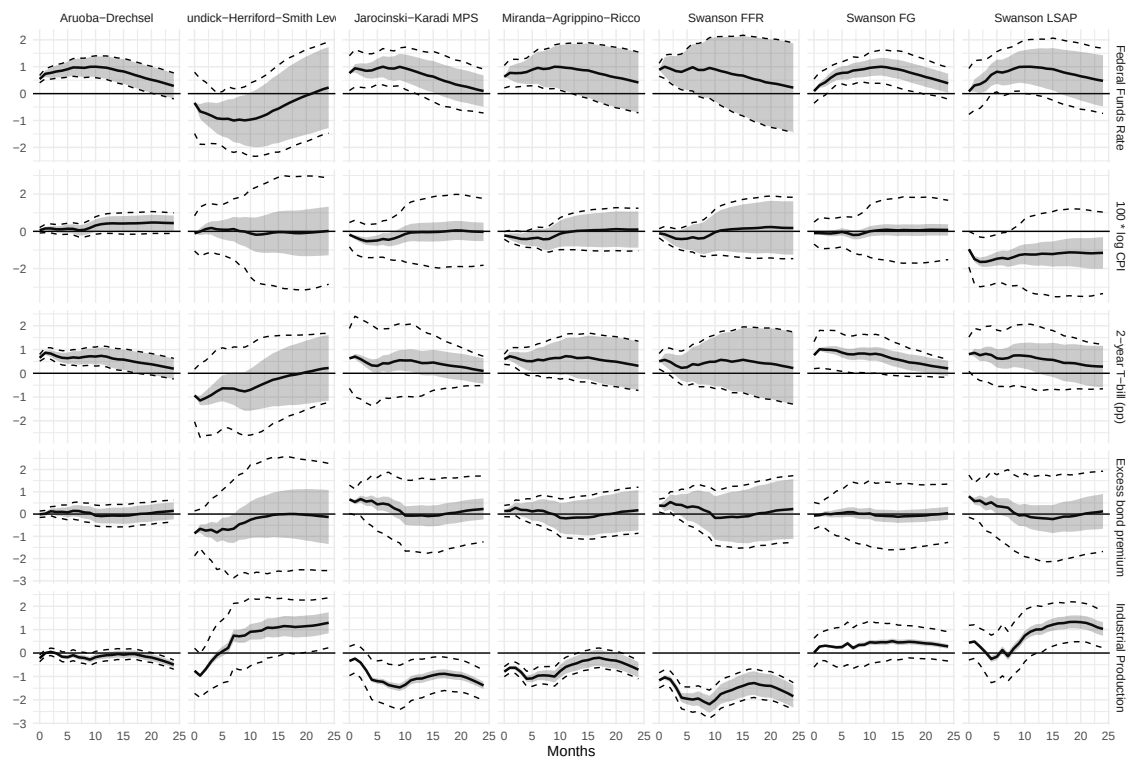


Figure 13: Impulse Responses to Empirical MPS (Largest Set)

Figure shows impulse responses from a VAR to each of the underlying empirical MPS. Each column shows the response to a different shock. The shaded region indicates the 90% confidence interval conditional on the impact values; the dotted lines indicate the 90 % confidence intervals accounting for impact uncertainty. The VAR lag length is 11 and is chosen by AIC.

shock) and the Phillips curve (a “supply” shock). We also simulate two instruments for each of the two non-monetary shocks, one strong and one weak. We then assess our method against these data in two distinct exercises: a long-sample assessment using 25,000 periods of simulated data; and a short-sample assessment using fewer periods.

Since the aim here is to test our method, rather than shock identification per se, we assume that the target empirical monetary shock is noisy (so $\text{var}\xi_t^j > 0$) and inherently monetary in nature, but that it has a non-trivial term structure (so $\beta_h^j \neq 0$ for some $h > 0$). Then the subsequent exercises answer the question: if a shock were identified perfectly, under what conditions would we correctly measure its term structure?

Parameter	Interpretation	Value	Parameter	Interpretation	Value
$1 - \alpha$	Returns to scale	0.67	β	Utility discount factor	0.997
ρ_a	Technology shock persistence	0.98	ρ_r	Interest rate smoothing in Taylor rule	0.49
ρ_z	Demand shock persistence	0.56	ρ_{zpc}	Supply shock persistence	0.49
σ	Risk aversion	1.0	φ	Labor supply elasticity	1.0
ϕ_π	Taylor rule inflation coefficient	1.5	ϕ_π	Taylor rule output gap coefficient	0.125
θ	Calvo parameter	0.75	Demand elasticity	6.0	
			$\{\beta_i\}_{i=0}^{L-1}$	EMPS lag structure	1
L	EMPS lag length	4	$\text{var}\xi$	MPS measurement error variance	0.003 ²
$\text{var}\nu_0$	EMPS surprise variance	0.003 ²	$\text{var}\nu_1$	EMPS 1-period news variance	0.0008 ²
$\text{var}\nu_3$	EMPS 1-period news variance	0.0001 ²	$\text{var}\nu_4$	EMPS 1-period news variance	0.00003 ²

Table 9: Parameters of the Monte Carlo Simulation

Table shows parameters used for simulations drawn from a calibrated version of the standard three-line New Keynesian model of Galí (2008).

F.1 Large Sample Properties

Table 10 reports the results of this first exercise, using standard parameters for the Taylor rule, an arbitrary declining term structure for the monetary policy shock, γ_i , and an AR(1) residual. The model parameters otherwise match a standard monthly calibration (see Table 9 for details). The specification in column (1) uses all four instruments, including the two strong ones. In this case, estimation recovers the correct parameters almost perfectly. With large enough sample and strong enough instruments, our method works.

In columns (2)-(3) we consider alternatives when multiple strong instruments are not available. In column (2), the case where we have only instruments for the demand shock, but one is strong.²² Although the point estimates are not as accurate as in the case with strong instruments for multiple shocks, the standard errors are appropriately wider, in that the true value of the parameters lies within two standard errors in all cases. In column (3) we include *only* weak instruments, although since one is a supply shock and one a demand

²²We need to use at least two instruments to estimate a Taylor rule with two contemporaneous variables. With fewer instruments than endogenous variables, the fitted endogenous regressors in the second stage are colinear.

shock there they cover more dimensions of variation in the data. Unsurprisingly, with only weak instruments, performances is worse. However, in column (4) we allow for the most obvious practical fix, just including lags of the weak instruments. The intuition is that past shocks can have distinct effects on contemporaneous endogenous variables. In this case, adding just six lags results in improved performance for almost all point estimates, and confidence intervals which continue to nest the true parameters.

Model			Four-stage IV				OLS			
			(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Taylor Rule</i>	ϕ_π	1.500	1.501 (0.003)	1.566 (0.076)	1.524 (0.031)	1.519 (0.024)	1.417 (0.003)	1.405 (0.003)	1.404 (0.003)	1.404 (0.003)
	ϕ_y	0.125	0.128 (0.002)	0.120 (0.010)	0.146 (0.015)	0.137 (0.010)	0.093 (0.001)	0.084 (0.001)	0.084 (0.001)	0.084 (0.001)
<i>Term Structure</i>	γ_0	0.435	0.435 (0.005)	0.433 (0.006)	0.450 (0.014)	0.443 (0.010)	0.402 (0.004)	0.394 (0.004)	0.393 (0.004)	0.392 (0.004)
	γ_1	0.109	0.107 (0.004)	0.021 (0.254)	0.059 (0.052)	0.097 (0.017)	0.111 (0.004)	0.110 (0.004)	0.104 (0.004)	0.104 (0.004)
	γ_2	0.017	0.017 (0.004)	-0.054 (0.196)	-0.018 (0.039)	0.010 (0.013)	0.016 (0.004)	0.017 (0.004)	0.016 (0.004)	0.016 (0.004)
	γ_3	0.004	0.008 (0.004)	-0.028 (0.104)	-0.013 (0.022)	0.003 (0.008)	0.009 (0.004)	0.009 (0.004)	0.007 (0.004)	0.007 (0.004)
Residual lag length, L			1	1	1	1	1	1	6	24
Demand instrument			2	2	1	1				
Supply instruments			2	0	1	1				
Instrument lags			0	0	0	6				
F-test, first stage, π_t			374.2	7.7	26.2	7.3				
F-test, first stage, y_t			102.2	37.5	9.1	3.2				
Lagged y_t							No	Yes	Yes	Yes

Table 10: Monte Carlo Simulation: Long sample

Table reports the results of estimating the term structure from 25,000 periods of simulated data using a New Keynesian model with a well-identified monetary policy shock. Columns (1) to (4) use our four-stage IV approach, and columns (5) to (8) estimating the Taylor rule by OLS. In all cases where only one instrument is used, it is the weakest one available.

In columns (5) to (8) we repeat this exercise for OLS estimation. In all cases, OLS is inaccurate in large samples and provides misleadingly narrow confidence intervals. As is well-known, OLS estimates of Taylor rules are inconsistent, and so even including extensive lagged endogenous variables and interest rates as controls does not fix this problem.

In summary, our four-stage IV method works in large samples. It is best with strong instruments but including instruments lags in the first stage can help offset some of the problems of instrument weakness. Even the worse version of the IV approach is superior to the best OLS method.

F.2 Small Sample Properties

We now report the performance of our method on repeated small samples. To avoid a profusion of specifications, we select three models, a “strong instruments” IV, a “weak instruments” IV, and a long-lagged OLS. These versions, which correspond to columns (1), (4), and (8) of Table 10, represent reasonable best- and worst-cases for the IV and the best case for OLS in practice.

Figure 14 reports the distribution of the estimates for the Taylor rule coefficients, for repeated samples of 250, 500 (which is close to our sample size), and 750 observations. This shows that if strong instruments are available, then our four-stage method is unbiased and relatively powerful even in small samples. When instruments are weak, the method is much less powerful. In general, the distribution of OLS is tight, but it is biased and to a non-trivial extent in samples of size relevant to our work.

It is important to note that the results presented here do not contradict Carvalho et al. (2021). There, the authors argue that Taylor rule estimation by OLS is, in most reasonable cases, better than GMM using lagged endogenous variables. Although they are inconsistent, the small-sample bias of OLS estimates Taylor rule is proportional to the variance of endogenous variable due to the monetary shock. Since this is small, the resulting bias is also small. One intuition for their findings is that lagged-variable GMM addresses endogeneity but only by throwing out contemporaneous co-variation of endogenous variables and the policy rate (except for that due to autocorrelation). In contrast, OLS uses information about the current period, but at the price of endogeneity bias. The key result of Carvalho et al. (2021) is that this is usually a price worth paying. Our method gives the best of both worlds. It exploits contemporaneous variation in the endogenous variables, but by isolating only the variation due to non-monetary shocks it corrects for endogeneity bias. This also has important implications for some of the limitations of OLS laid out by Carvalho et al. (2021), who show that both OLS and lagged-variable GMM perform increasingly poorly for either persistent monetary shocks or for Taylor rule coefficients near to unity. Neither of these limitations apply to our method. And, as we will see in the next paragraph, these properties matter when estimating the term structure of monetary policy.

Figure 15 shows the same distribution of point estimates in the repeated samples for the estimated term structure. When assessed in this way, the four-stage IV estimator performs well even when instruments are weak, producing better estimates of the term structure compared to OLS *even* in cases where the IV Taylor rule estimates are clearly inferior. This is particularly true for γ_0 , arguably the most important entry in the term structure. The intuition is that because the term structure of the EMPS is the projection of the whitened estimated Taylor rule residuals onto the EMPS, and because the endogenous variables are autocorrelated, the estimated residuals are correlated with the whitening regressors. As a

result, the relationship between $\hat{\phi}$ and $\hat{\gamma}$ is effectively concave.²³ The confidently incorrect Taylor rule estimates from OLS are heavily penalized by this convexity and so are projected onto term structure estimates far from the truth. In contrast, the IV estimates are more spread out and so the mapping to $\hat{\gamma}$ is, on average, more forgiving.

Another important measure of the accuracy of test statistics is the coverage ratio. To assess this, we compute for each simulation the p-value of a hypothesis test with the true null. If the distribution of these p-values is uniform, then the test will have good coverage ratios at all confidence intervals. Note that this is a joint test of both the point estimate and its variance. From a practical perspective is the gold standard for creating useful estimators: if an estimate delivers uniform p-values, it says that one can do reasonably accurate inference about the data generating process, *even if the point estimates are inaccurate*.

Figure 16 reports the distribution of p-values for the estimated Taylor rule coefficients. Throughout, the IV estimates are relatively close to the diagonal. Performance is better, of course, when instruments are strong or when the sample size is larger. But for samples similar to the size we use, the results are generally good. In contrast, confidence intervals based on OLS cannot be trusted for any parameters for small sample sizes or for ϕ_π, ϕ_y at any sample size.

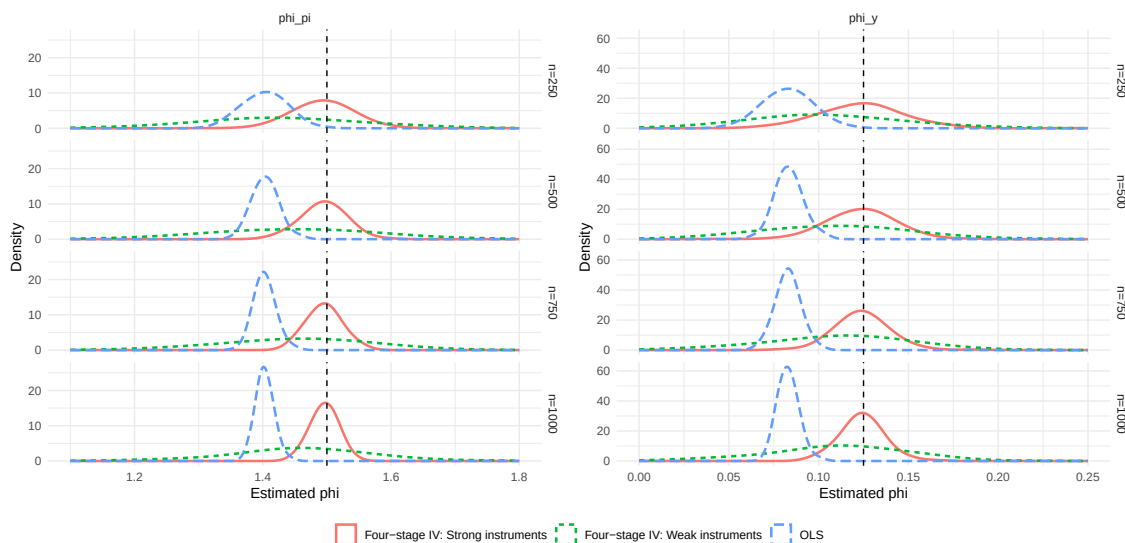


Figure 14: Short sample simulation: Distribution of estimated Taylor Rule coefficients

Figure shows the distribution of the estimated Taylor rule coefficients at different sample sizes. Calculations drawn from disjoint subsamples of a 100,000 period simulation.

²³This concavity can be seen in the slight skews for the smallest sample sizes in Figure 15.

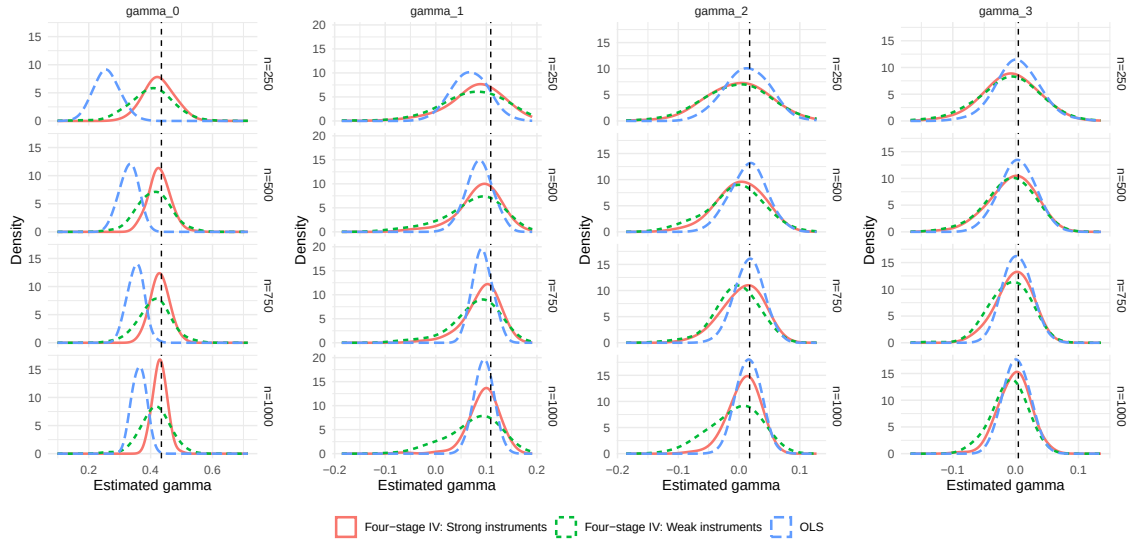


Figure 15: Short sample simulation: Distribution of estimated term structure of monetary policy.

Figure shows the distribution of the estimated monetary policy term structure at different sample sizes. Calculations drawn from disjoint subsamples of a 20,000 period simulation.

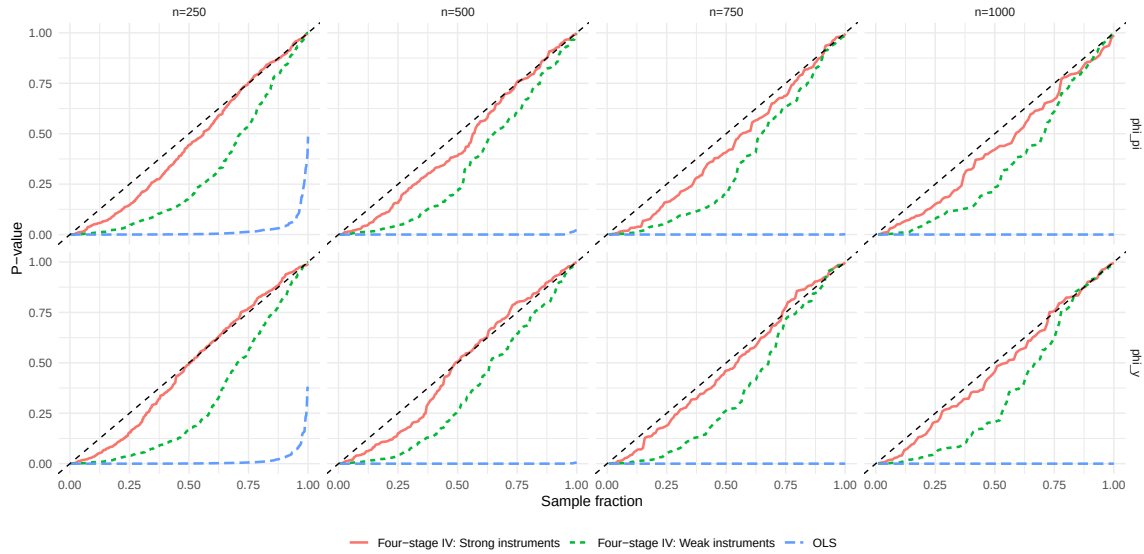


Figure 16: Short sample simulation: Distribution of p-values for Taylor Rule coefficients

Figure shows the distribution of the p-value of the true model parameter for Taylor rule coefficients at different sample sizes, using small-sample point estimates and standard errors. Perfect coverage ratios would produce diagonal lines. Calculations drawn from disjoint subsamples of a 20,000 period simulation.